



BASIC CEMENTING TRAINING

Cement Slurry Design, Equipment & Operation



Jakarta, July 4th 2026

Speaker

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**THINK
WORK
STAY**
SAFE
Teamwork · Innovation · Excellence

01 CONSEQUENCE A POOR CEMENTING

02 WHAT'S CEMENTING

03 CEMENTING EQUIPMENT

04 SLURRY

05 CASE STUDY

06 CEMENTING CHALLENGES



Part 1

Consequence of A Poor Cementing

The Criticality of a Cementing Job

Cementing is a **one-shot** job!

The consequence of a poor cementing job is **expensive**
and can be **disastrous!**

A Movie Based on a True Story



DEEPWATER HORIZON

-----The "911" in the history of oil industry

The BP blowout in Gulf of Mexico

April 2010

Reality is much worse than the movie!

Damage to Environment



**4.9 million barrels' oil
spilled. 2,500 KM² of
water covered in oil!**

The Suffering of the Wild Life



Financial Loss of the Company



BP was fined \$20.8 billion!

Investigation of the Deepwater Horizon Accident

The investigation demonstrated that the blowout was caused from a poor cement job!



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Panel Says Firms Knew of Cement Flaws Before Spill

By JOHN M. BRODER
Published: October 28, 2010

WASHINGTON — [Halliburton](#) officials knew weeks before the [fatal explosion](#) of the [BP](#) well in the Gulf of Mexico that the cement mixture they planned to use to seal the bottom of the well was unstable but still went ahead with the job, the presidential commission investigating the accident said on Thursday.

[Enlarge This Image](#)



Gerald Herbert/Associated Press

The Deepwater Horizon oil rig is seen burning in Venice, La. Tests performed before the deadly blowout of BP's oil well in the Gulf of Mexico should have raised doubts about the cement used to seal the well, but the company and its cementing contractor used it anyway.

Green
Blog posts about the

In the first official finding of responsibility for the blowout, which killed 11 workers and led to the [biggest offshore oil spill](#) in American history, the commission staff determined that Halliburton had conducted three laboratory tests that indicated that the cement mixture did not meet industry standards.

The result of at least one of those tests was given on March 8 to BP, which failed to act upon it, the panel's lead investigator, Fred H. Bartlit Jr., said in a [letter](#) delivered to the commissioners on Thursday. "There is no indication that Halliburton highlighted to BP the significance of the foam stability data or that BP personnel raised any questions about it," Mr. Bartlit said in his report.

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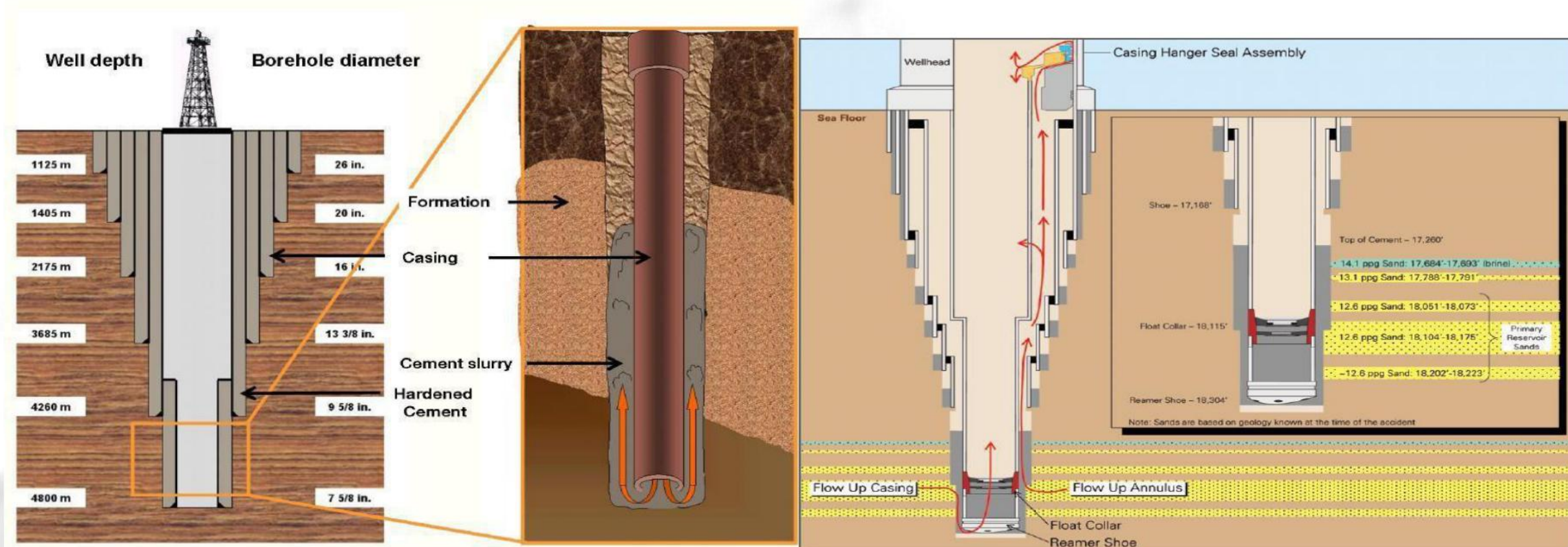
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Reasons for the Blowout

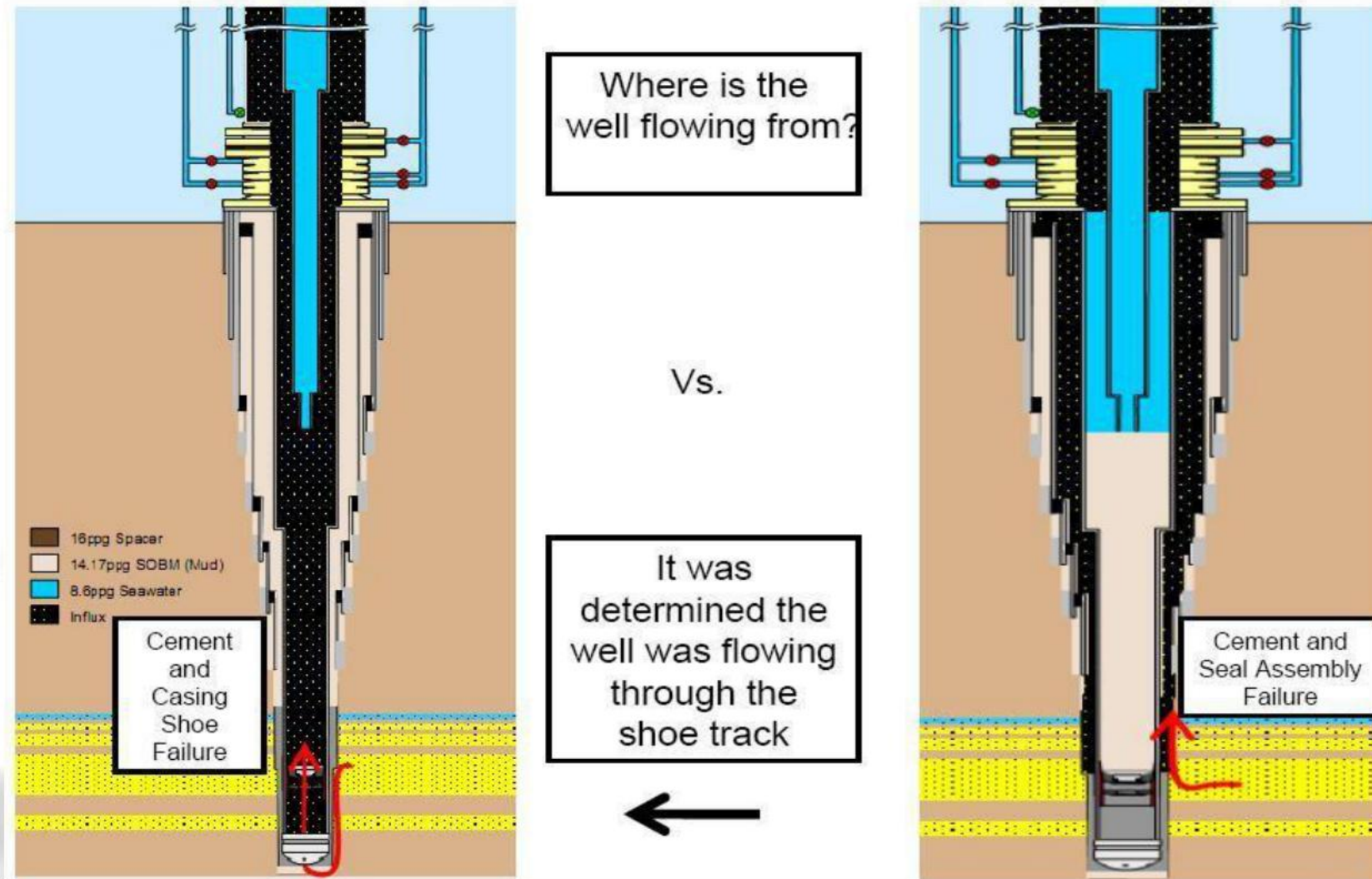
Direct reasons led to the blowout:

The cement failed to effectively seal the casing annulus after cementing.

-- According to a Joint Survey by the U.S. Bureau of Ocean Energy Management, Oversight and Administration and the Coast Guard.



Failure Mechanisms led to the Accident



The Role & Responsibility of Cementing Engineers!



It is the responsibility of cementing team to successfully finish each cementing operation.

Let “DEEPWATER HORIZON” only happened in the movie.

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04 SLURRY

05 CASE STUDY

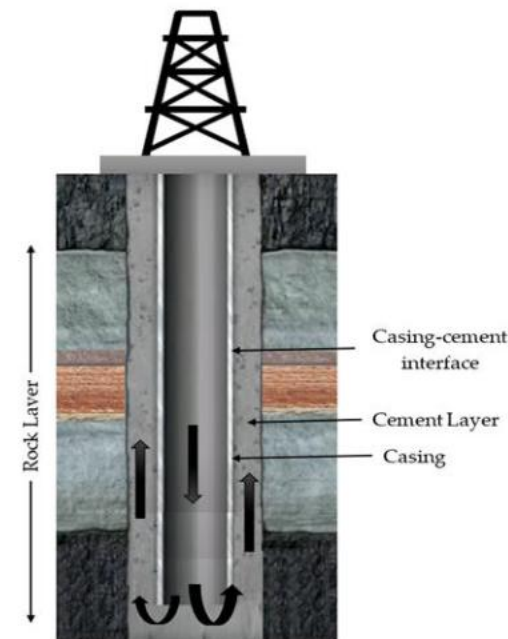
06 CEMENTING CHALLENGES



1. Introduction

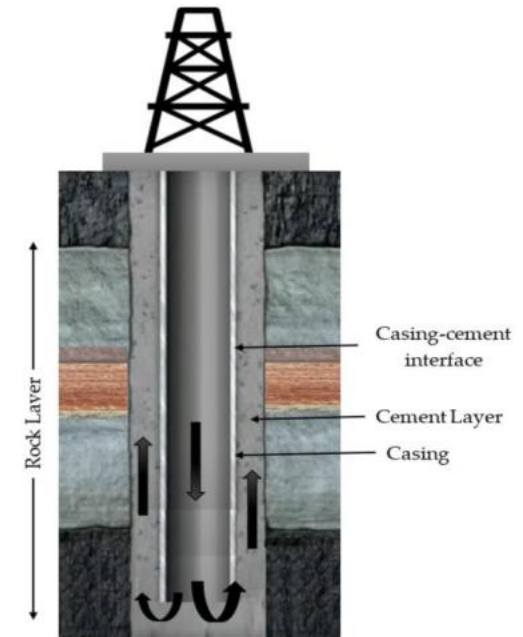
- ◆ The introduction of a cementitious material or cement slurries into the annulus between casing and open hole or previous casing or both to:
 - Provide zonal isolation
 - Support axial load of casing strings and strings to be run later
 - Provide casing support and protection
 - Support the borehole
- ◆ It is a **key part** of the drilling and completion operation.
- ◆ A conventional cementing job includes two processes: **running the casing/liner string** and **placing the cement slurry in place**.

“Cementing is an one-shot job! ”

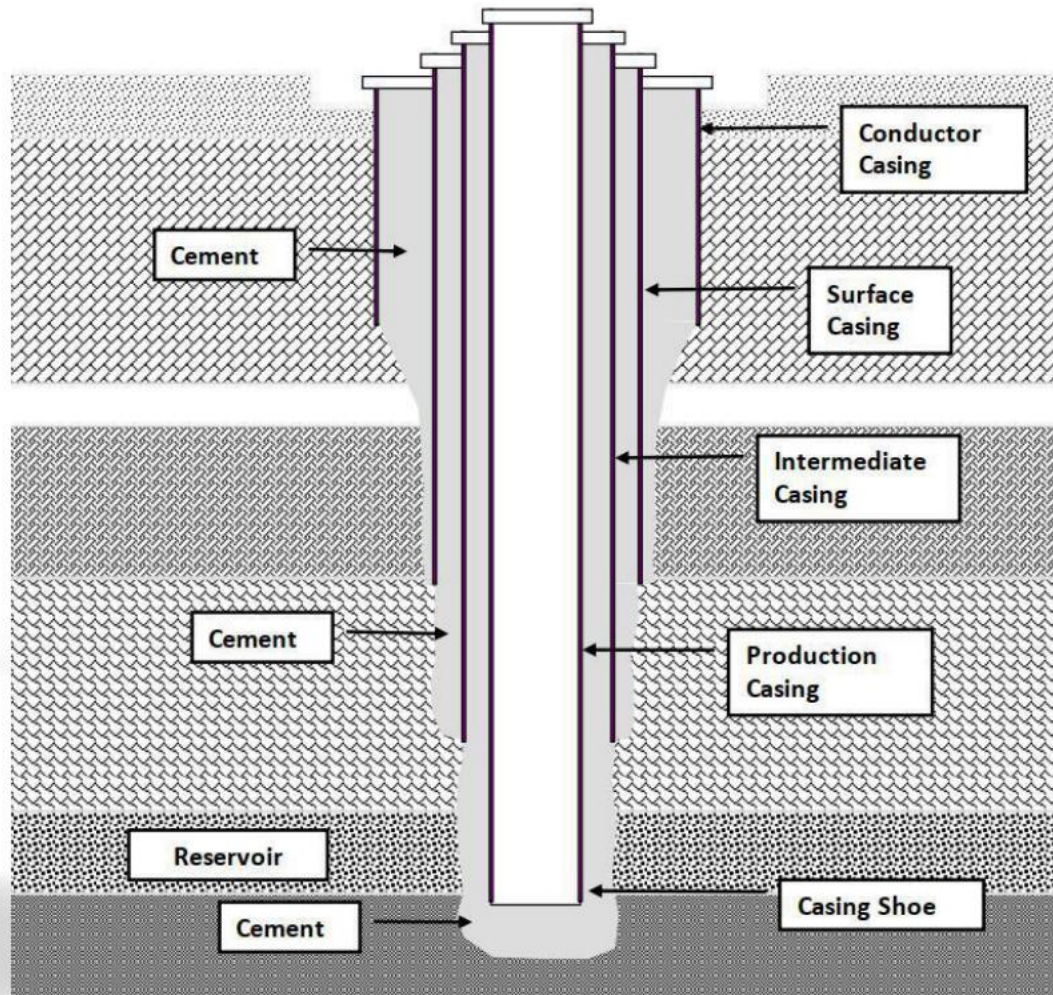


2. Objectives of Cementing

- ◆ Isolate and consolidate formations with risks of hole collapse or downhole losses.
- ◆ Provide the solid basis for the casing for the installation of the wellhead equipment.
- ◆ Isolate and seal off the different oil, gas and water layers, and prevent channeling of different layers, provide good conditions for the well test and production of oil and gas.
- ◆ Protect the fresh water in the upper formations from being contaminated by oil, gas, brine and other fluids in the lower part of the formations.
- ◆ During the development stage, provide good conditions for stimulation operations like acidizing and fracturing.



“Cementing is an important job! ”



Well structure

1. Conductor pipe
2. Surface casing
3. Intermediate casing
4. Production Casing

3. Classification of Casing Strings

Conductor Pipe

Conductor is usually **the first** and **shortest** casing string. Its main functions are:

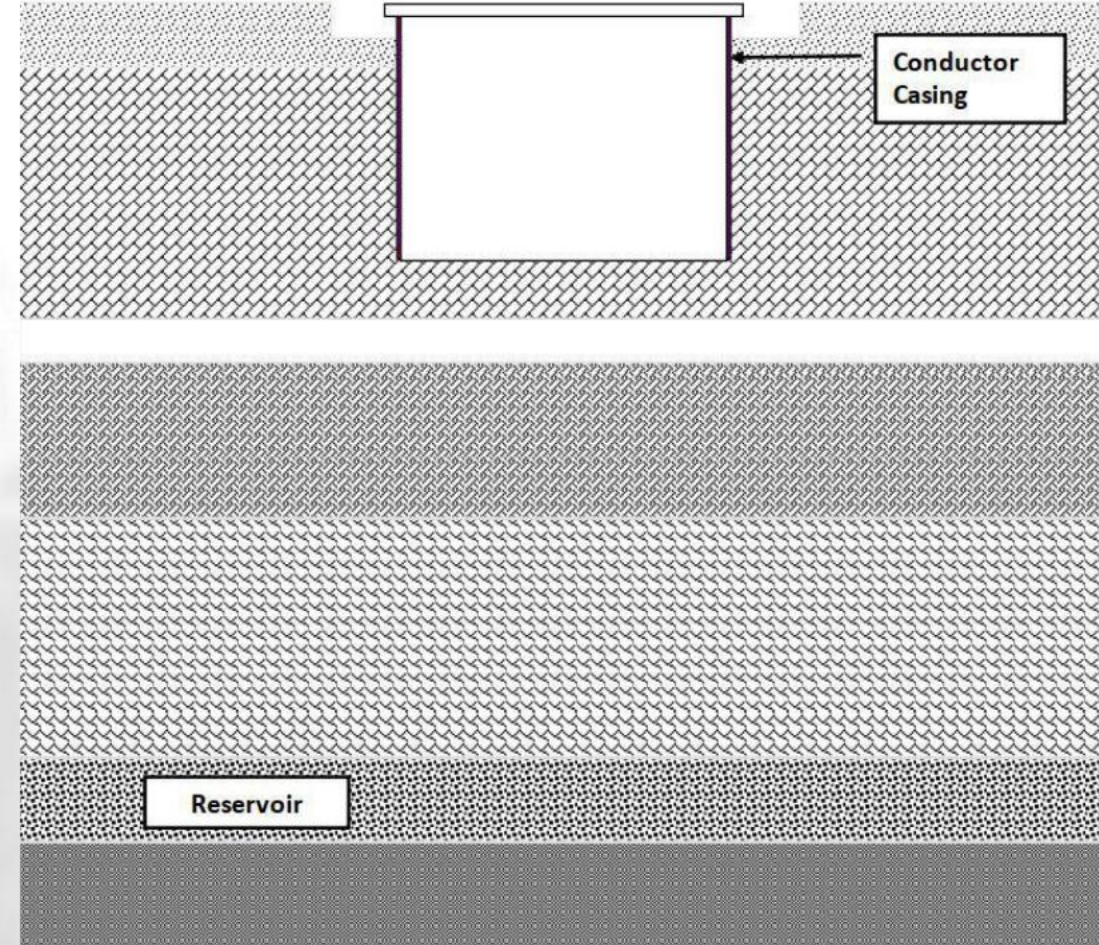
- ◆ Prevents washing out under the rig
- ◆ Provides elevation for flow nipple
- ◆ Protects shallow sands from contamination by drilling mud

Challenges:

- ◆ Possible occurrence of shallow water flows

Others:

- ◆ Large excess
- ◆ Stab-in cementing common
- ◆ Accelerated neat cement



3. Classification of Casing Strings

Conductor Pipe

A. Stab in Cementing Method

Key Points :

1. Less cement contamination
2. Less channelling
3. Small displacement volume
4. Pump until cement to surface
5. Less job time (rig time)
6. Less cement

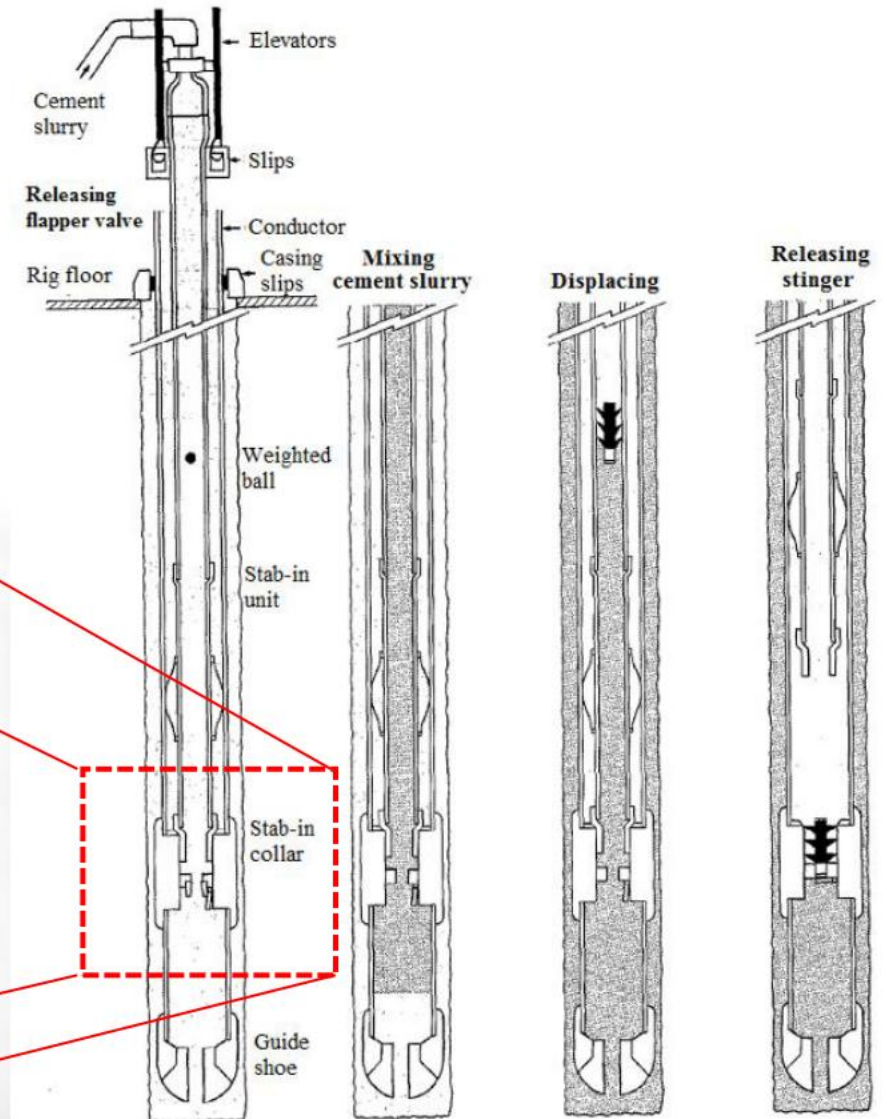
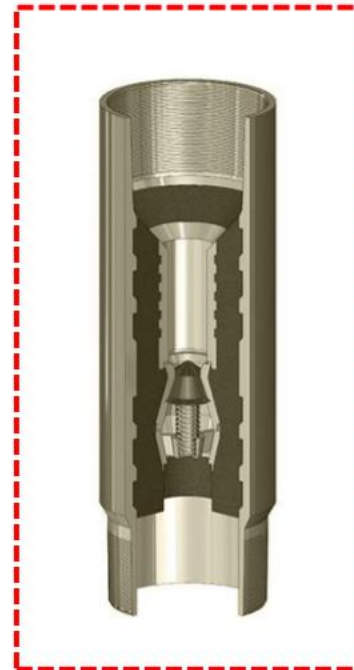


FIGURE 9: Inner string cementing operation (Nelson, 1990)

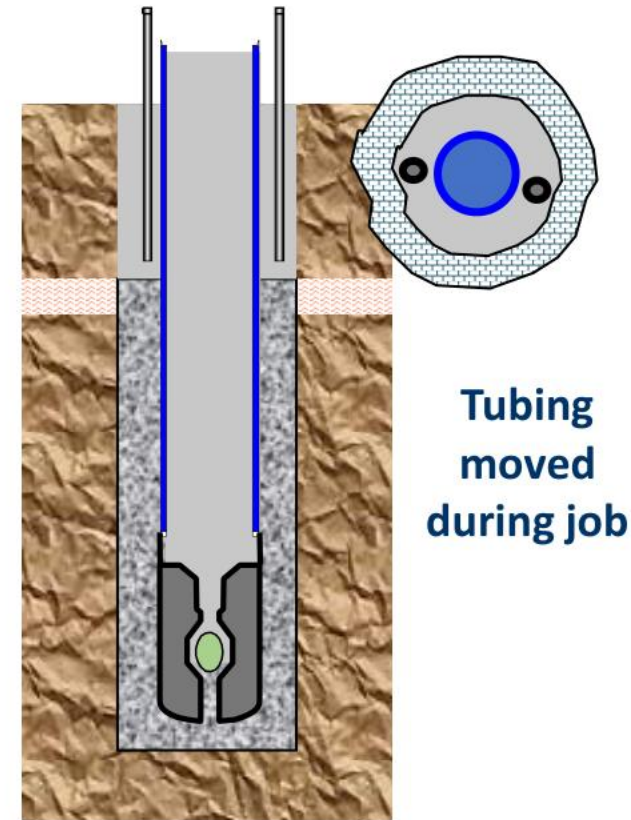
3. Classification of Casing Strings

Conductor Pipe

B. Macaroni Pipe Cementing Method (Top Job)

Key Points :

1. Bring cement to surface
2. Macaroni tubing used
3. Max. depth 250-300 ft
4. High friction pressures
5. Non-standard connections



3. Classification of Casing Strings

Surface Casing

Surface casing is the **second string** of casing, it usually

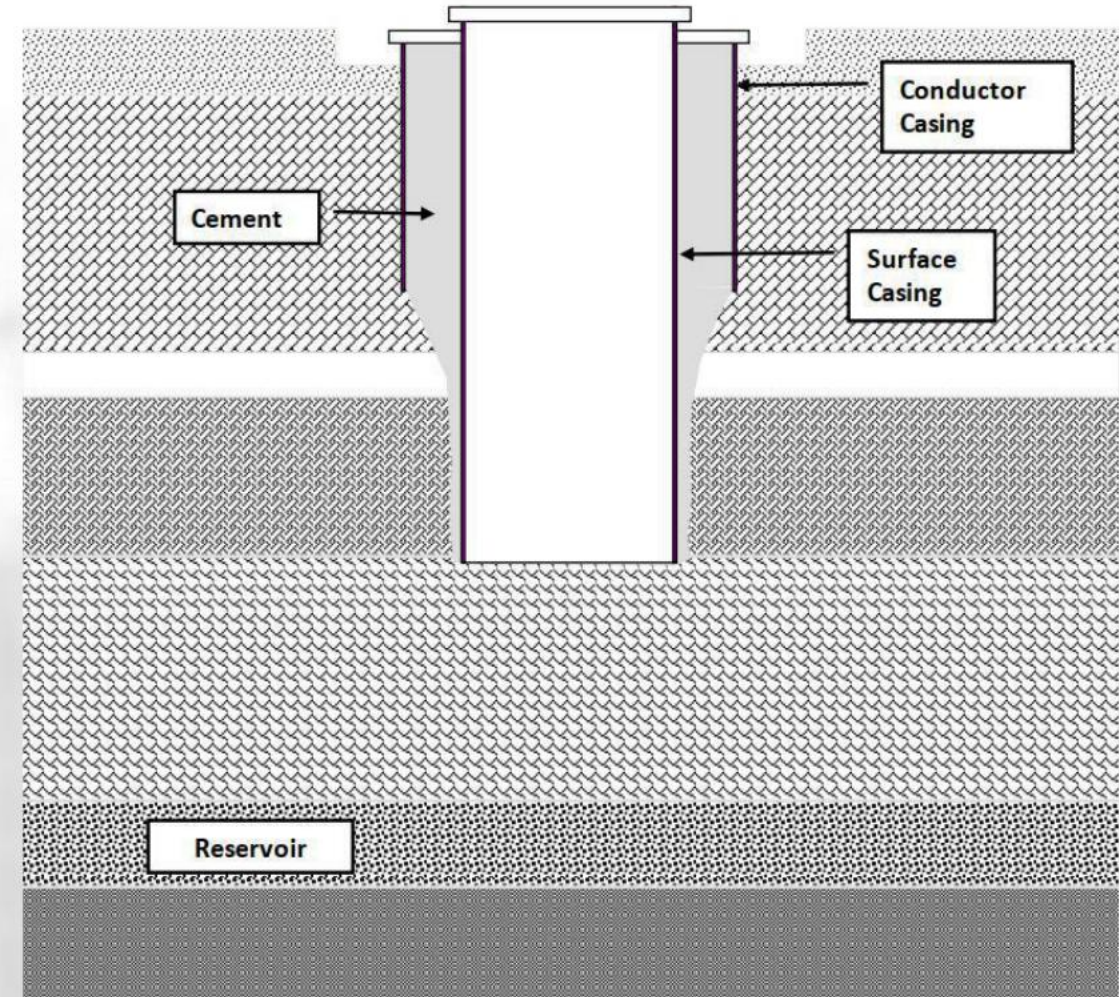
- ◆ Protect surface fresh-water formation
- ◆ Case off unconsolidated or loss areas
- ◆ Provide a competent mechanical base for subsequent operations (BOP, casings, etc.)

Challenges:

- ◆ Possible occurrence of shallow water flows
- ◆ Low temperatures (offshore)
- ◆ Drilling through gas hydrates (offshore)

Others:

- ◆ Light weight lead and neat tail slurries
- ◆ Large excess (50 - 150 %)
- ◆ Two plug method



3. Classification of Casing Strings

Surface Casing

Single Stage Cementing

Key Points :

1. Cementing head used
2. Top & bottom plug used
3. Shoe track
4. Lot of contamination

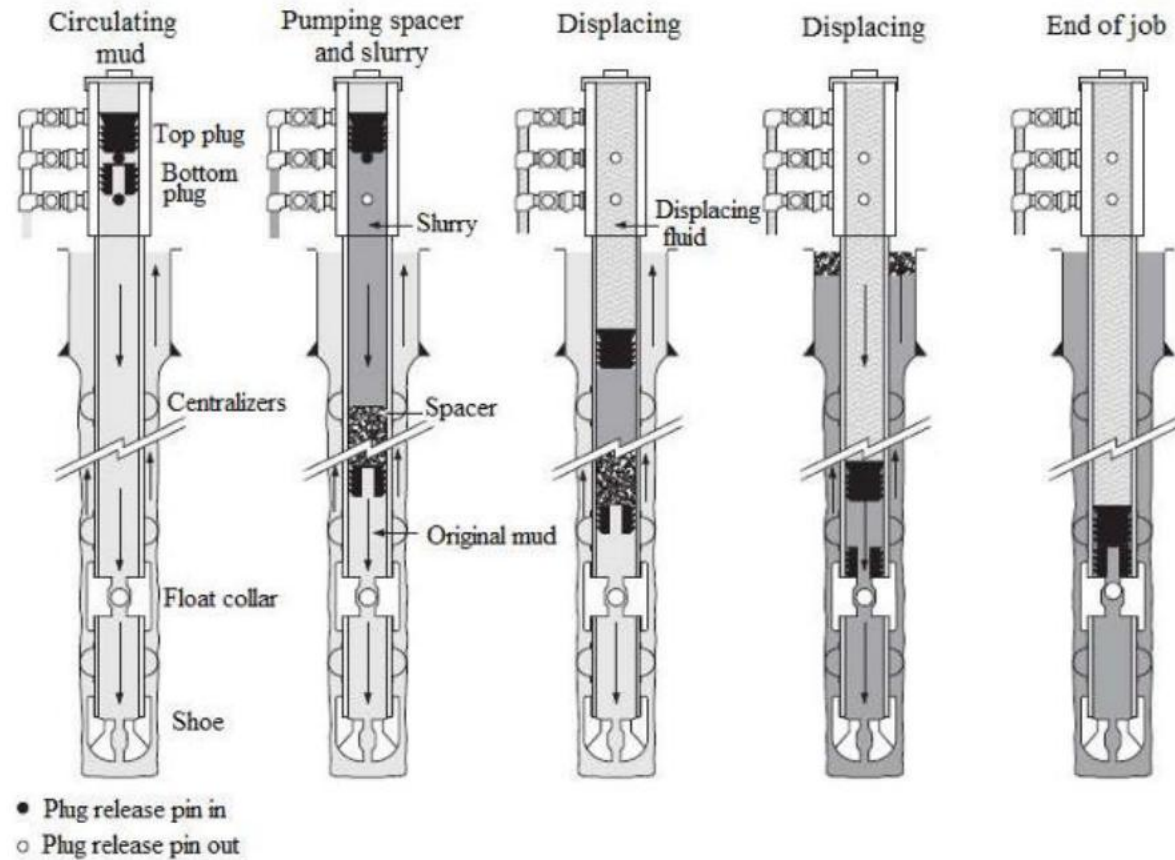


FIGURE 2: Single stage cementing operation (Nelson, 1990)

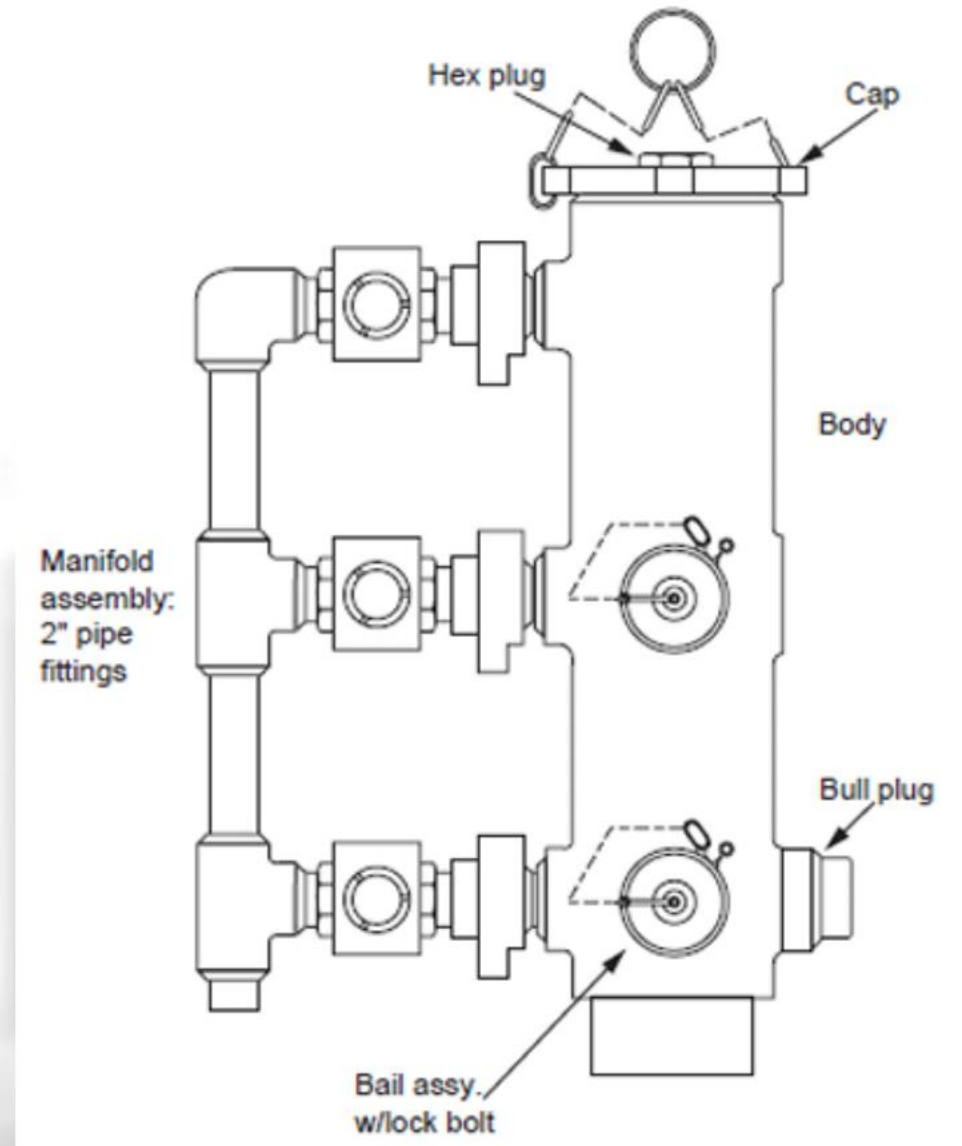
3. Classification of Casing Strings

Surface Casing

Cementing Head

Key Points :

1. Single plug or double plug type
2. Plug release indicator



3. Classification of Casing Strings

Intermediate Casing/Liner

Intermediate casing or intermediate liner is the casing string running between surface casing and production casing. It usually:

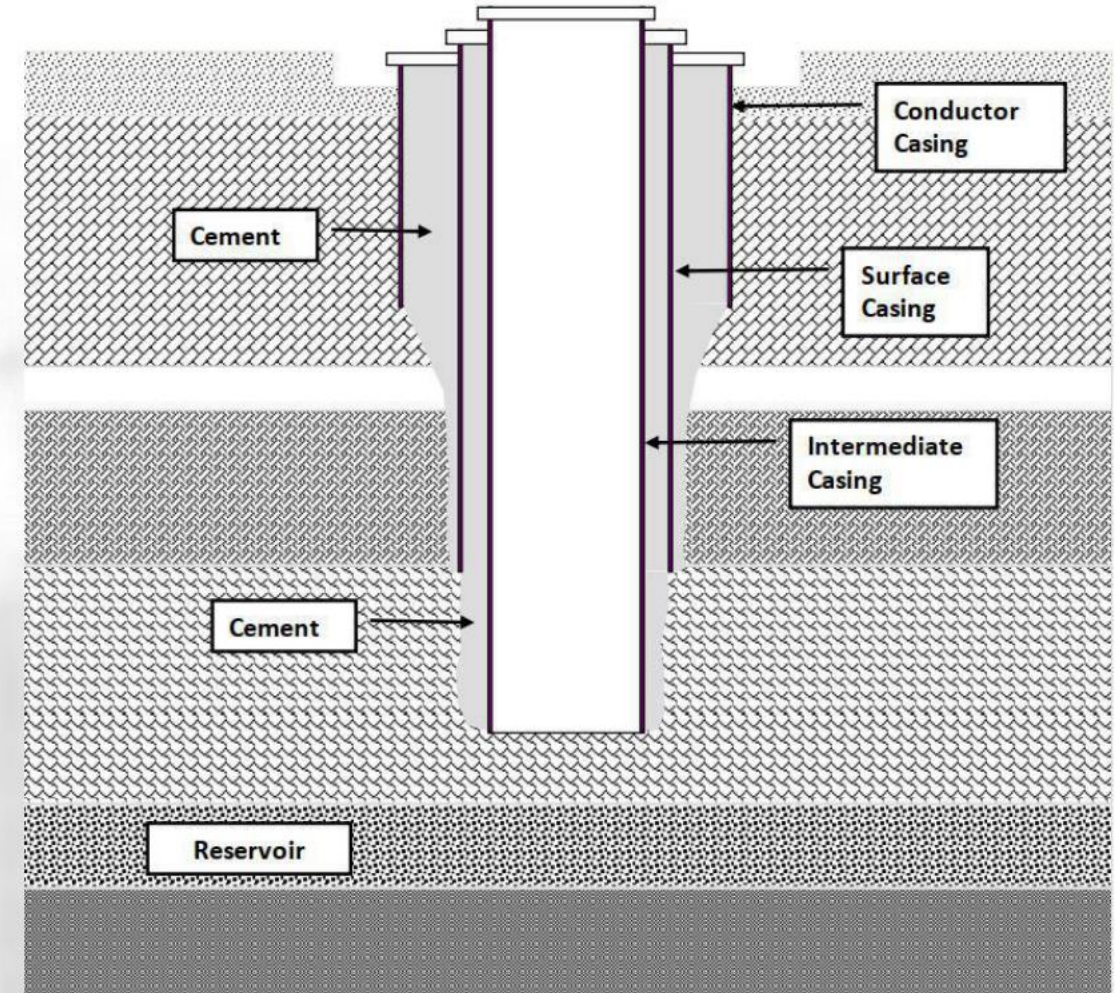
- ◆ Isolate hole into workable sections
- ◆ Seal off nonproductive, high-pressure zones above the zone of interest.

Challenges:

- ◆ Potential problems: over-pressured, loss zones, salt formations or heaving shales
- ◆ Narrow pressure window, between pore @ bottom & frac @ top

Others:

- ◆ Often need a two-stage job
- ◆ Best cementing practices are required
- ◆ Cemented to surface or to previous casing shoe
- ◆ Specialized slurries (light, heavy, salt etc)



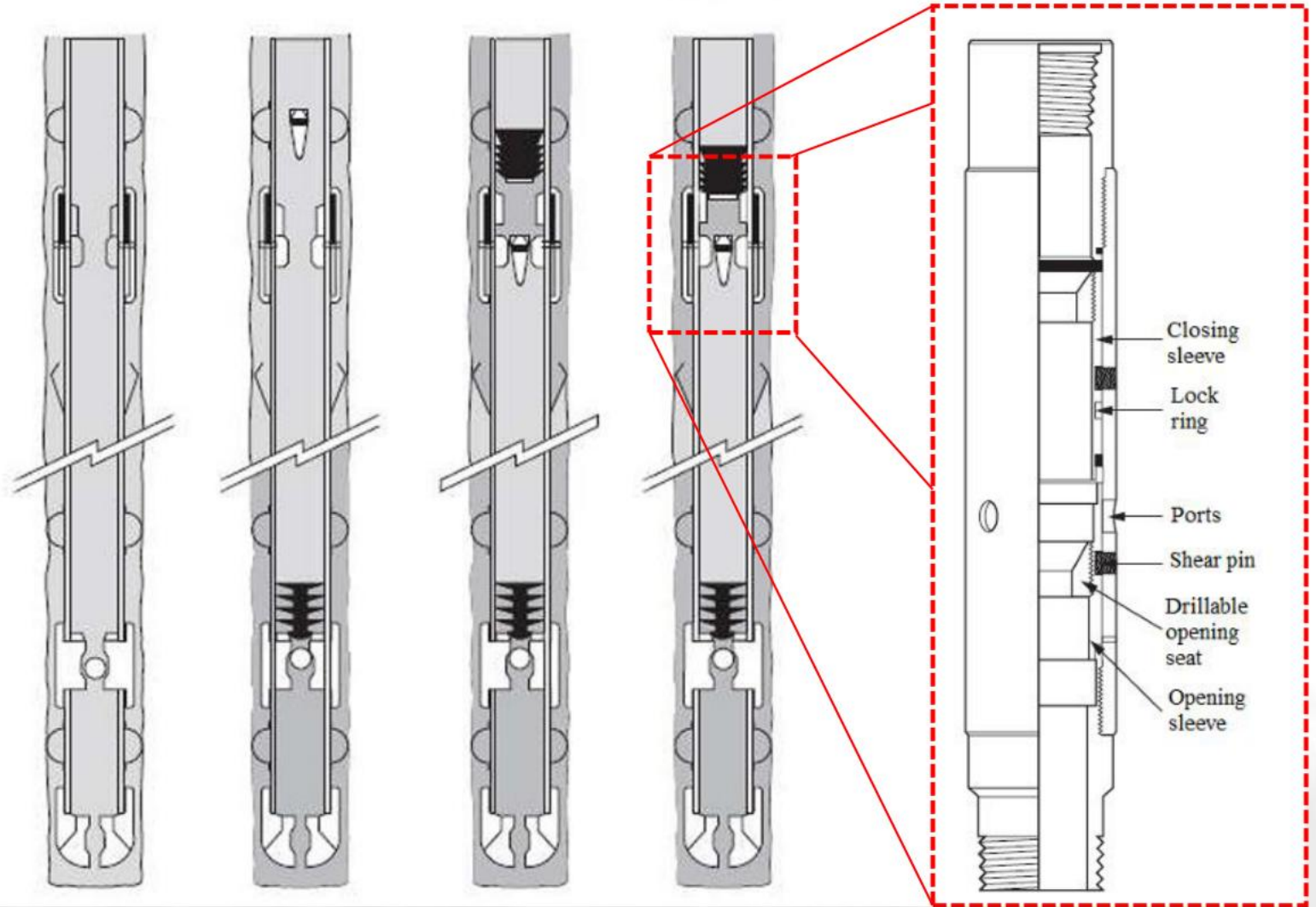
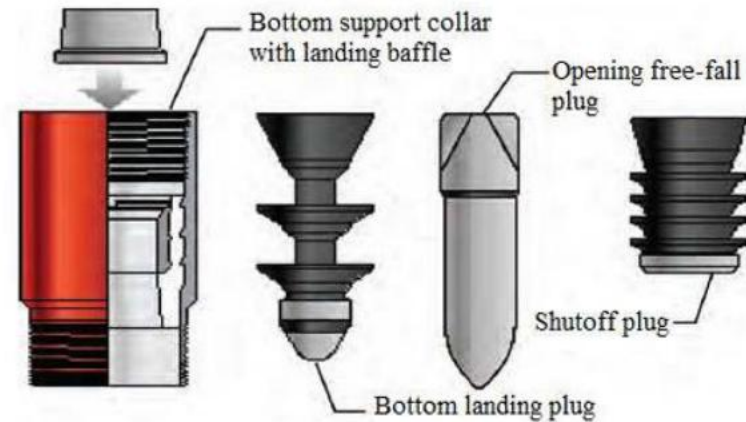
3. Classification of Casing Strings

Intermediate Casing/Liner

Dual Stage Cementing

Key Points :

1. Separation and isolation of zones
2. Reduces hydrostatic
3. Can leave zone in the annulus uncemented (cement at TD and surface)

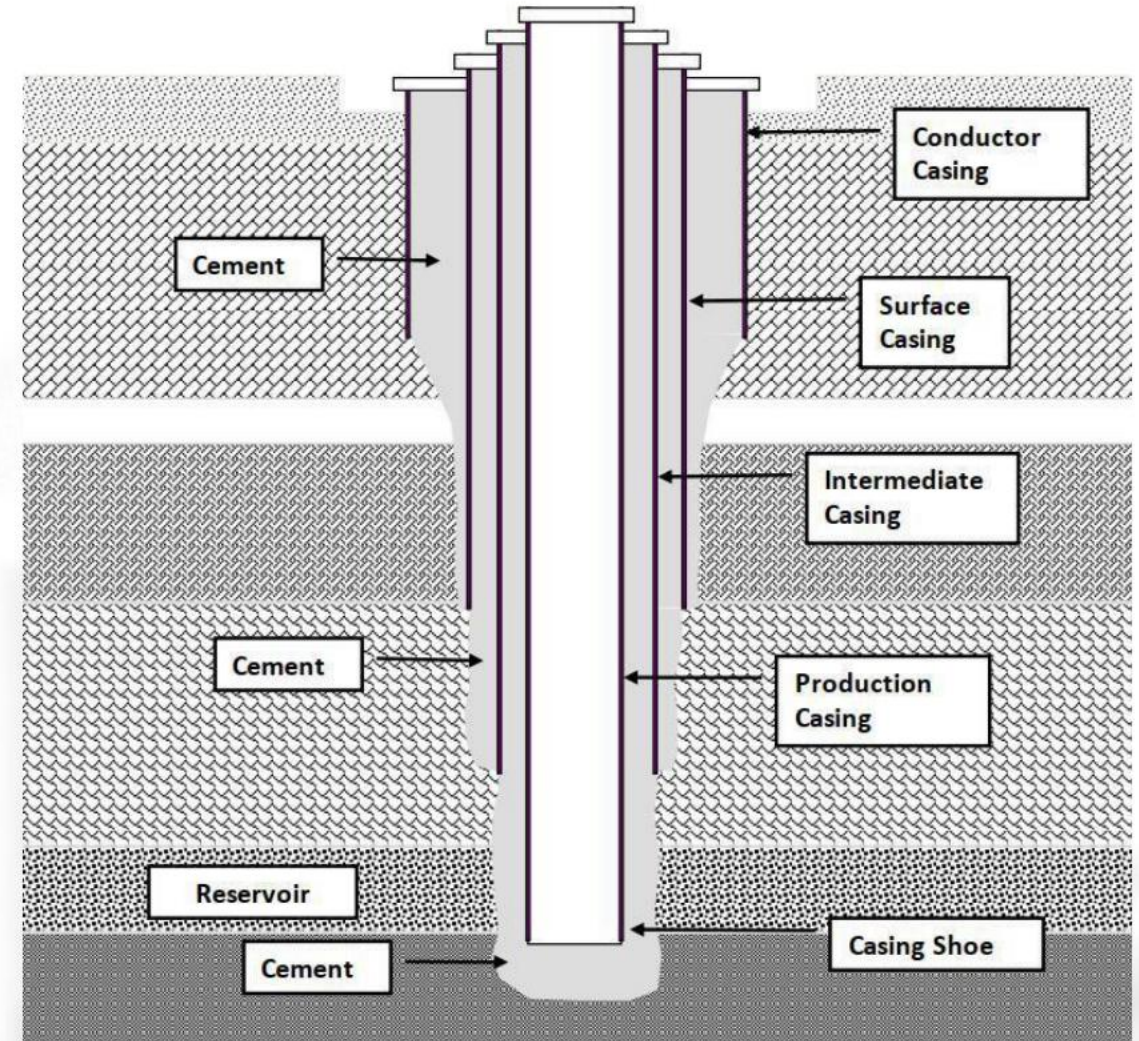


3. Classification of Casing Strings

Production Casing/Liner

Production casing or production liner can be the last tubular element cemented in the well.

- ◆ It isolates different zones above and within the production zone.
- ◆ Withstands all the anticipated loads during testing operations and production throughout the well's life.
- ◆ Any casing or liner that creates an annular space with the production tubing is treated as a production casing or liner.



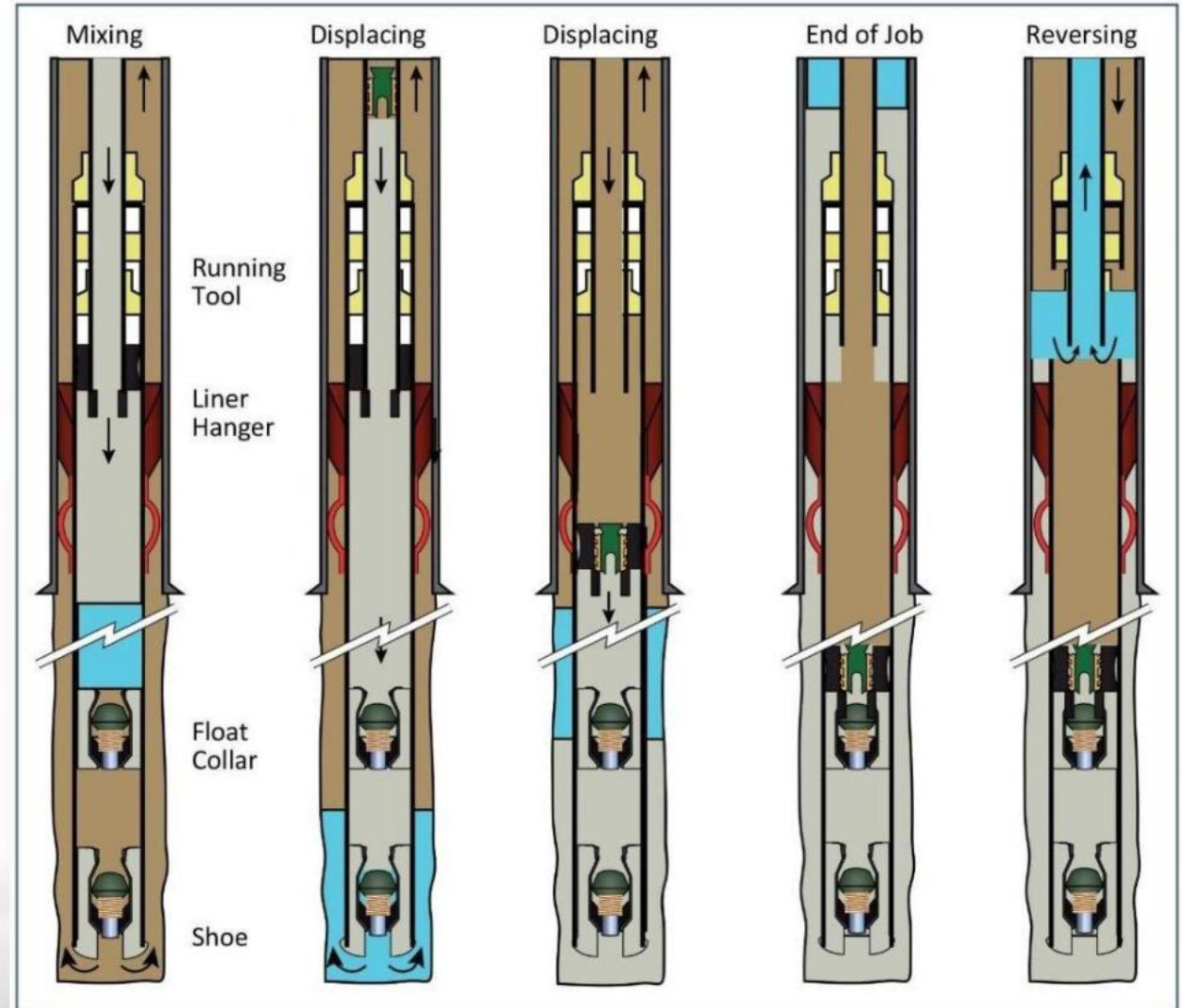
3. Classification of Casing Strings

Production Casing/Liner

Liner Cementing

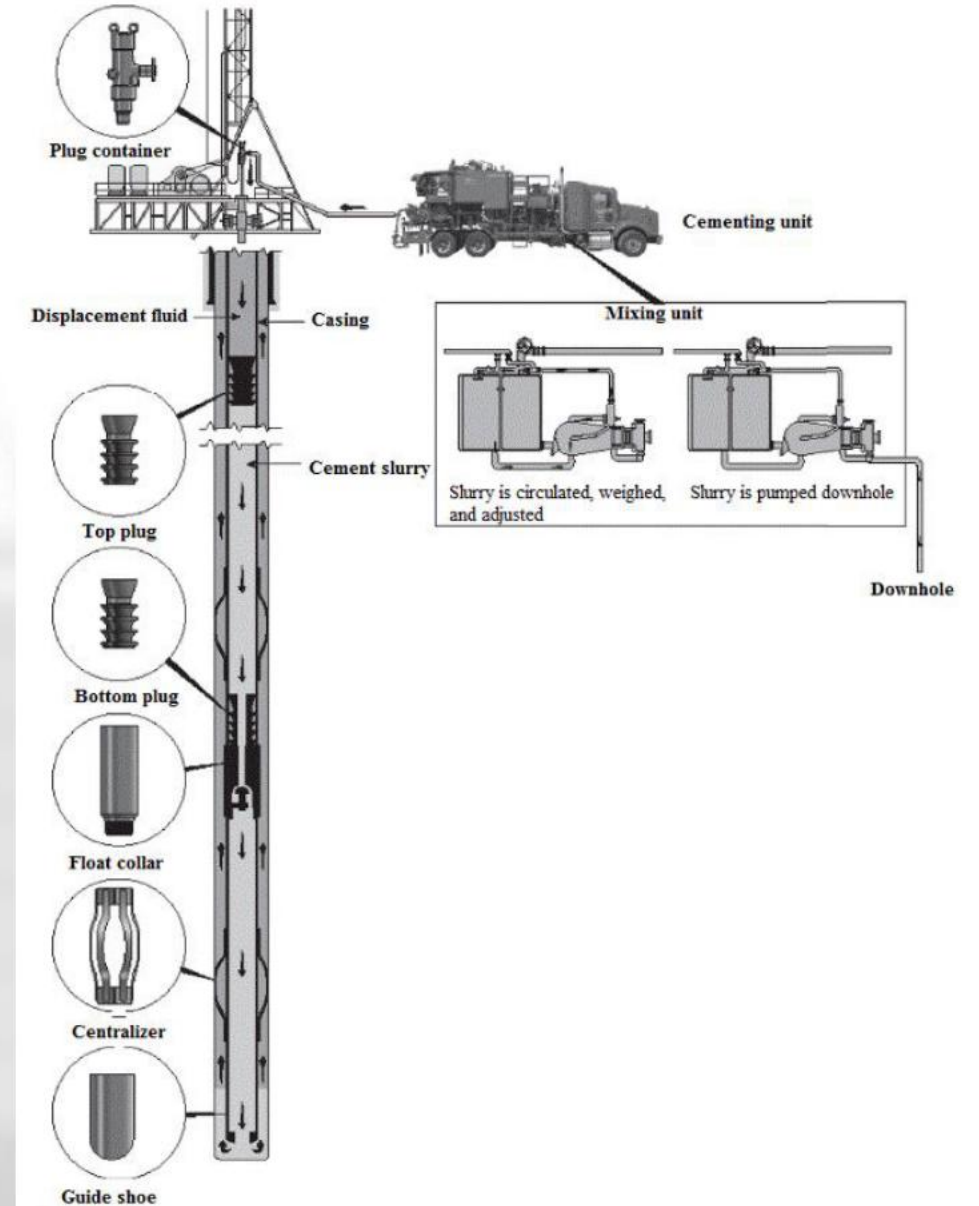
Key Points :

1. Requires less casing
2. Deeper wells
3. Small annular clearance
4. Specialized equipment



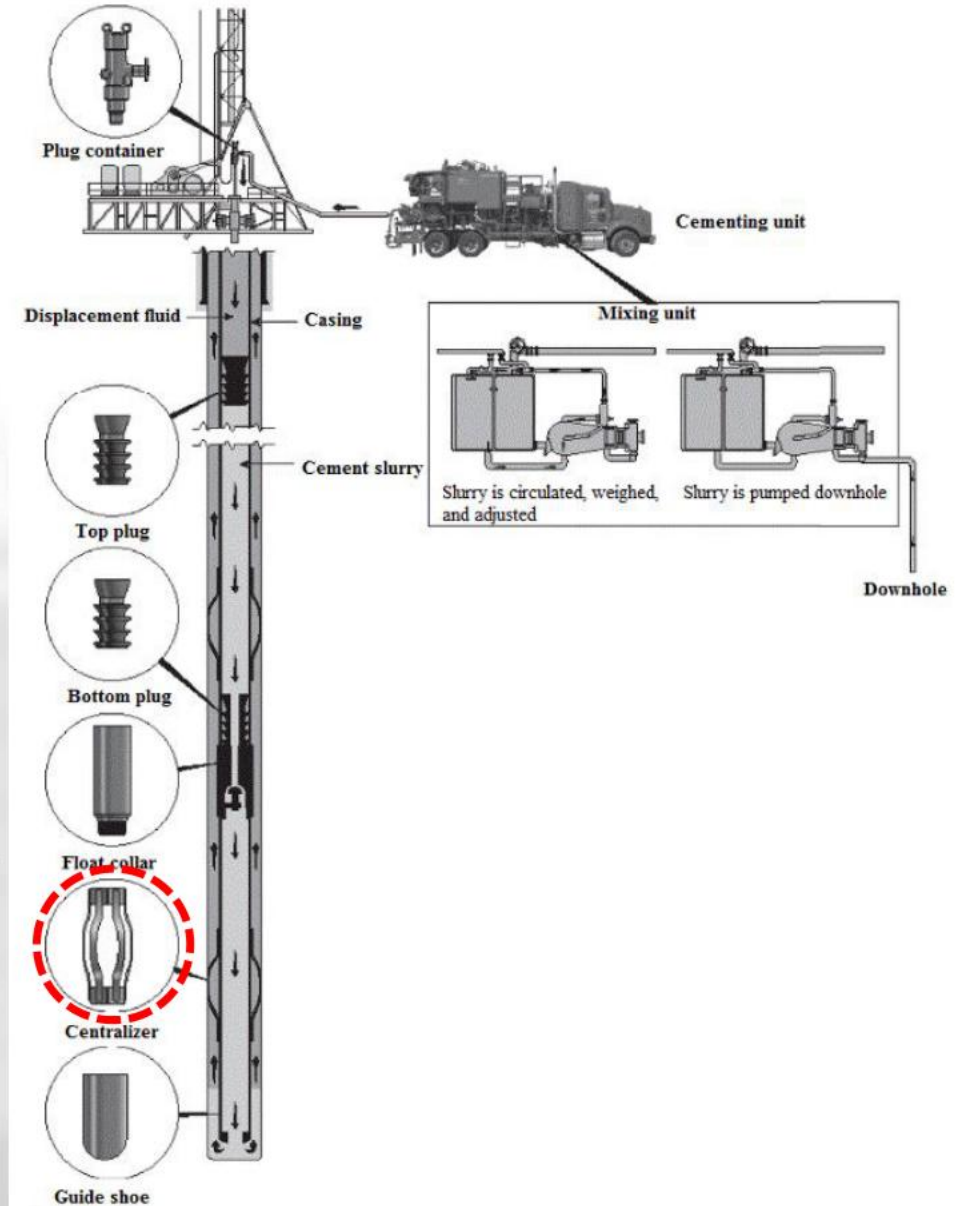
4. Casing Accessories

1. Centralizer & Stop Collar
2. Float Collar & Float Shoe
3. Wiper Plug

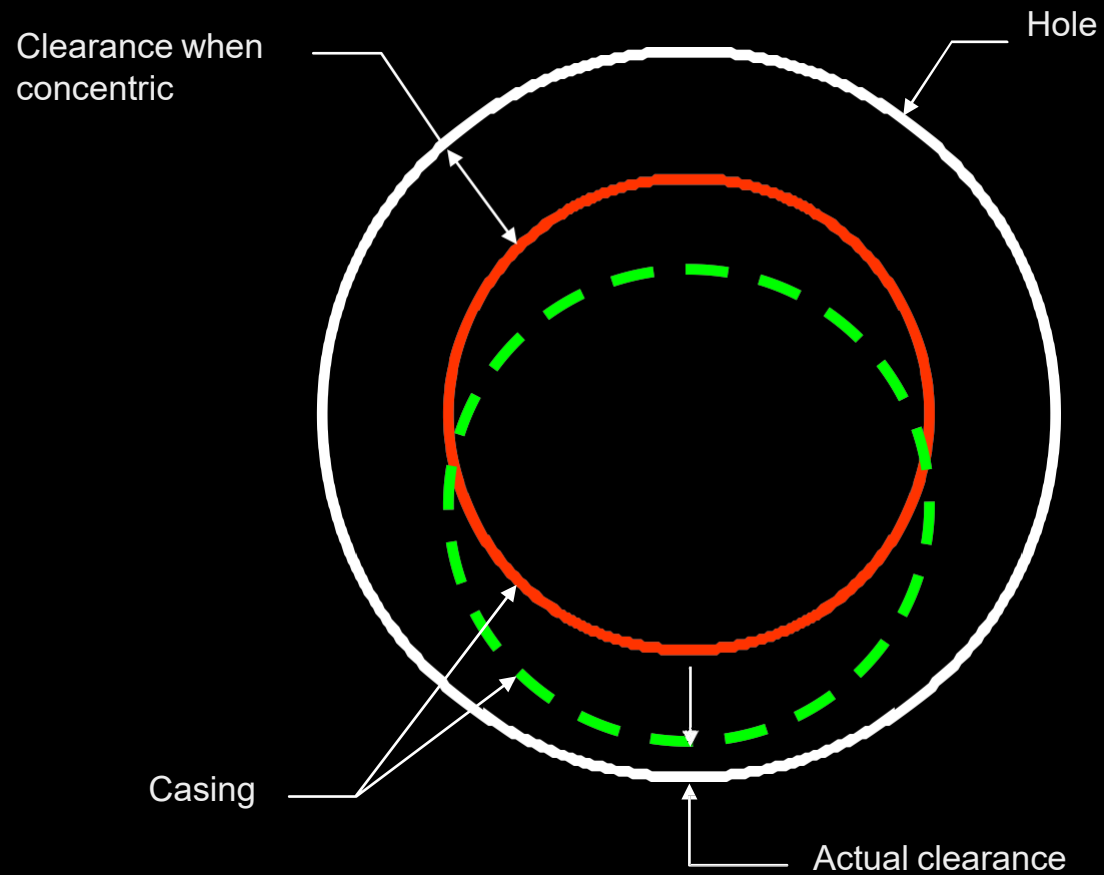


4. Casing Accessories

1. Centralizer & Stop Collar : Ensure the casing is centered against the wellbore walls, maximizing cement fill and preventing excessive annular space

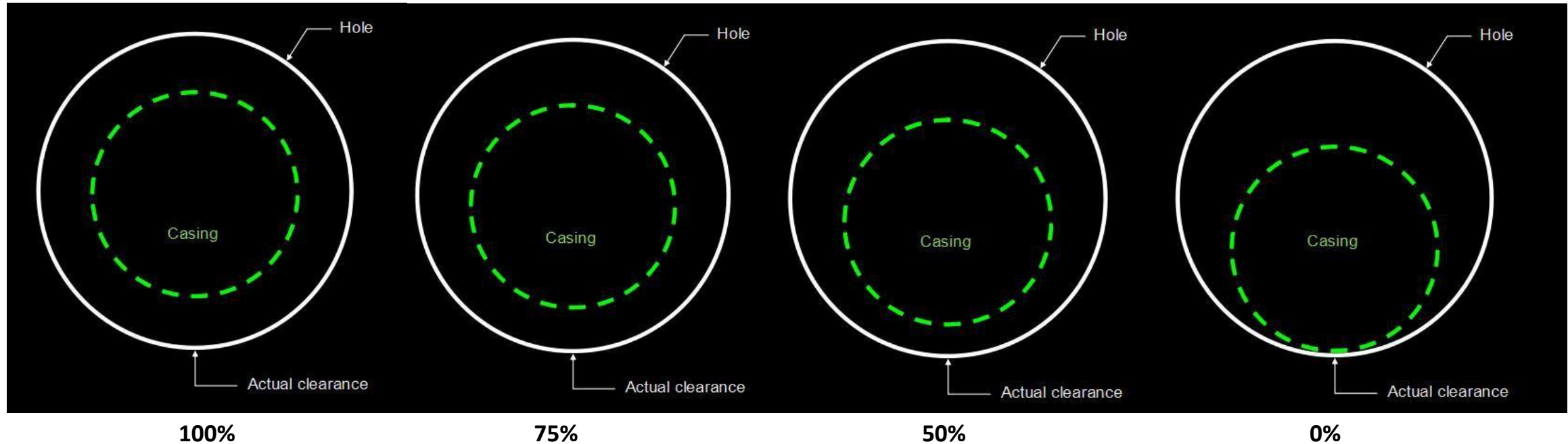


Centralizers - Standoff



$$\text{Standoff} = \frac{\text{Actual clearance}}{\text{Concentric clearance}} \times 100\%$$

Centralizers - Standoff



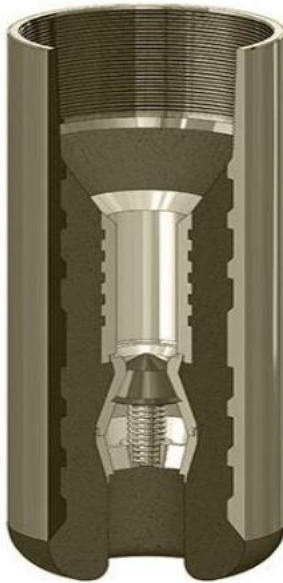
- ◆ API Spec 10D defines the minimum requirement for casing standoff as 67%. Enough centralizers should be utilised to meet the minimum standoff of 67%.
- ◆ Though a minimum 67% standoff has been set by API, 67% standoff will not guarantee adequate cement displacement efficiency in field operations.
- ◆ It is recommended to **get the BEST possible stand off** that can be achieved.

4. Casing Accessories

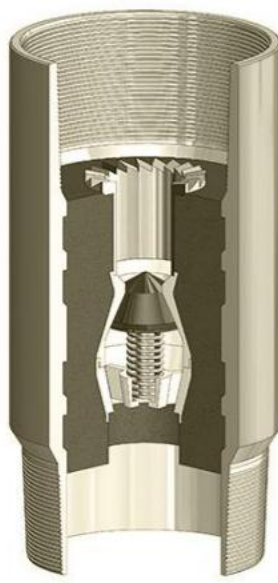
1. Centralizer & Stop Collar
2. Float Collar & Float Shoe : **Control the flow of cement during cementing, ensuring proper placement and preventing cement from being prematurely displaced**



Conventional
Float Shoe



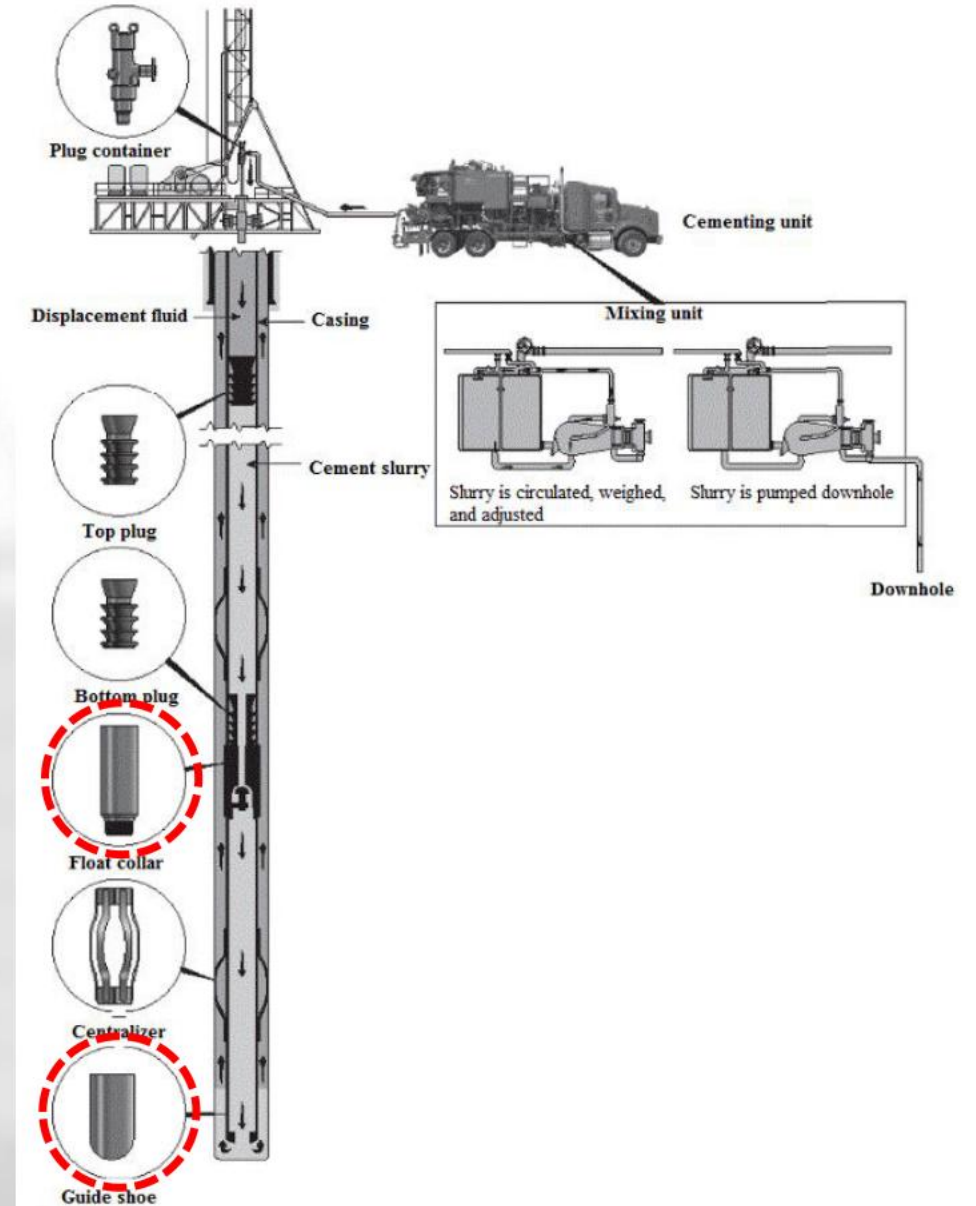
Stab-in Float
Shoe



Conventional
Float Collar



Stab-in Float
Collar



4. Casing Accessories

1. Centralizer & Stop Collar
2. Float Collar & Float Shoe
3. Wiper Plug : **Separate cement slurry from drilling fluids and displacement fluids, ensuring a clean and efficient cement job**



Non-Rotating
Bottom Plug



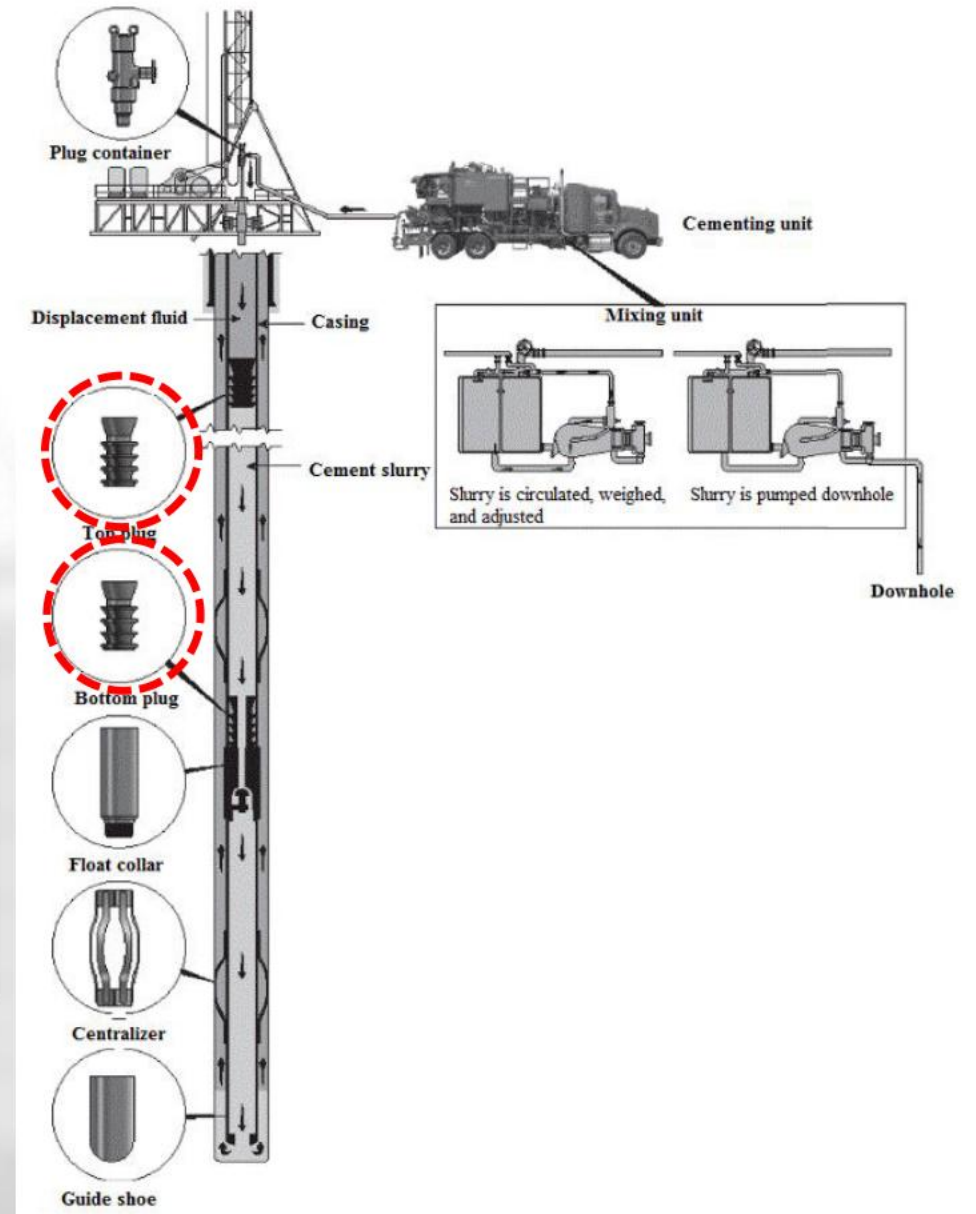
Non-Rotating
Top Plug



Conventional
Bottom Plug



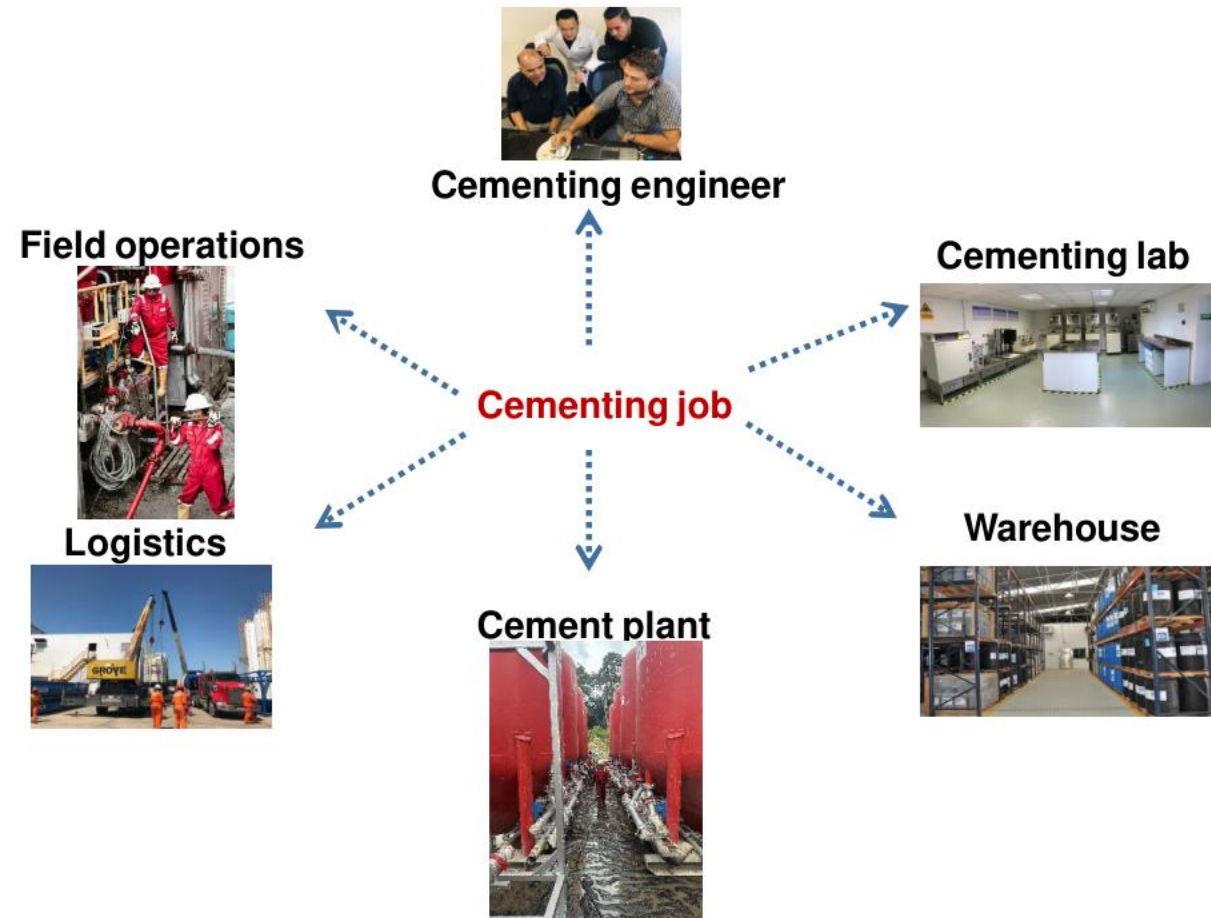
Conventional
Top Plug



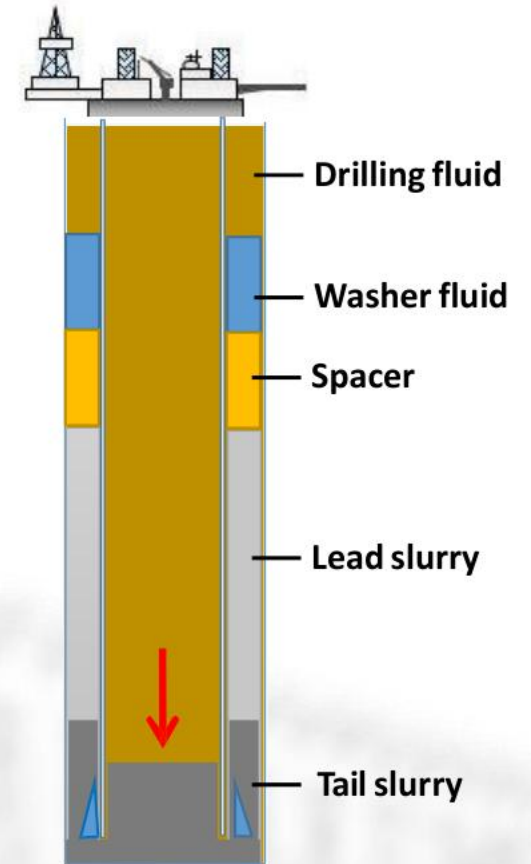
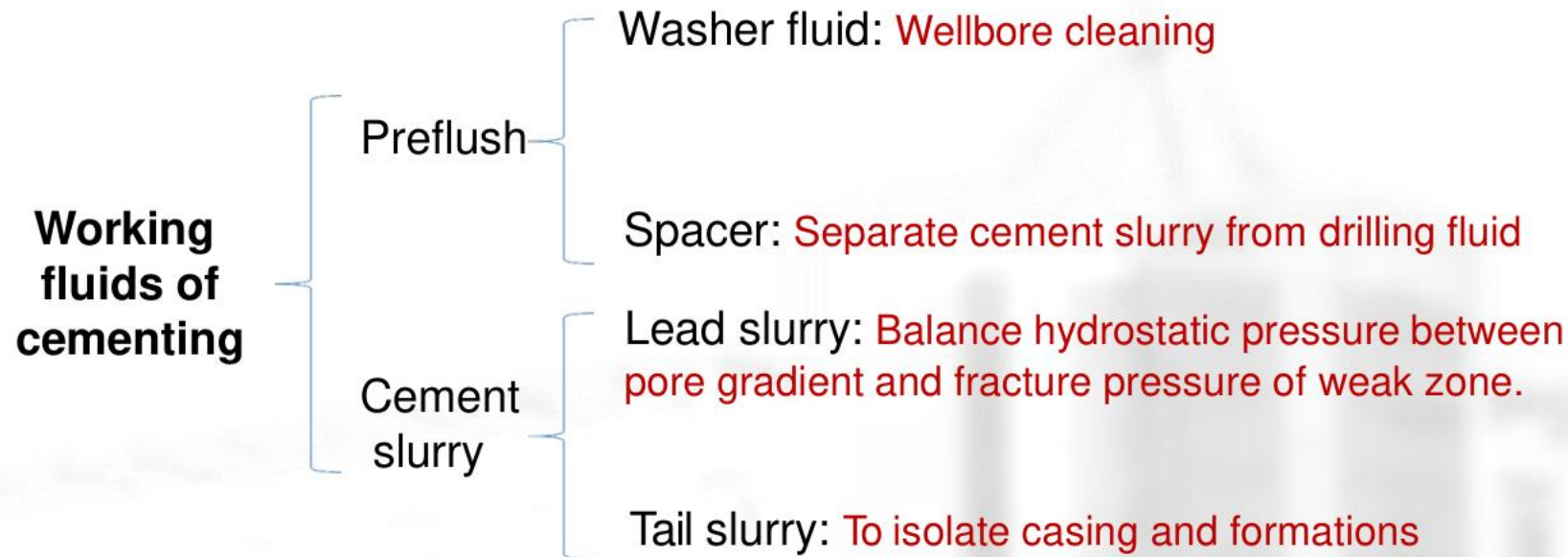
4. How Cementing Works?

Cementing is a systematic engineering. The completion of a good cementing job requires an integrated excellent collaboration of different disciplines.

1. Engineers collect information from client and discuss technical requirements of the cementing job. ↓
2. Engineers make cementing design. ↓
3. Cementing lab test slurry by the requirements of engineer and send report. ↓
4. Cement plant mix blend cement, warehouse prepare materials. ↓
5. Equipment, cement, additives be sent to the rig by logistic. ↓
6. Cementing crew finish the cement job.



5. Pumping the Working Fluids



Displacing process of cementing

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06 CEMENTING CHALLENGES



Overview

Cementing Operation: A multidisciplinary & An Integrated Team Work

Equipment & Tools

1. Cementing equipment
2. Rig tools
3. Accessories
4. Downhole tools
5.



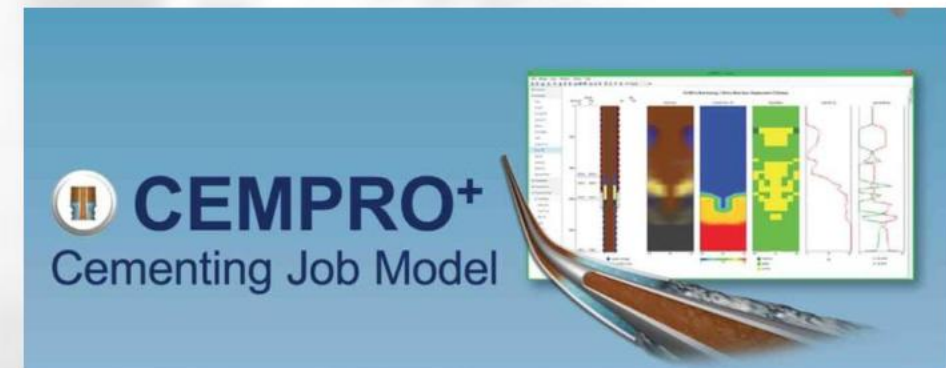
Chemistry of Cement

1. Cement slurry
2. Preflush
3. Additives
4. Reaction mechanisms
5.



Cementing Engineering

1. Fluid mechanics
2. Rock mechanics
3. Pipe string mechanics
4. Drilling engineering
5. Well logging engineering



Cementing Equipment

Cementing equipment is the equipment used to finish the cementing operations. Its main functions are:

- ◆ Storage, transfer, and mixing of cement and additives.
- ◆ Pumping cement slurry into the wellbore.



WATER SUPPLY SYSTEM

Provide clean water for mixing cement slurry



WATER TANK

Store and supply clean water for cement slurry

CEMENT SUPPLY SYSTEM

Store and deliver cement powder to the mixing system

CEMENT IN

Cement powder is delivered from bulk truck or cement bag



METERING TANK

Hold cement slurry before pumping to the wellhead

CEMENTING PUMP

Provide high pressure to pump cement slurry to the wellhead

DIESEL ENGINE

Provide power to drive the cementing pump

CEMENTING HEAD

Distribute cement slurry into the well through casing or liner

MIXING EQUIPMENT

Mix water and cement powder to produce cement slurry

Legend

- Water Line
- Cement Flow / Slurry Line
- High Pressure Slurry Line

4 x Systems of Cementing Equipment

1. Cement Supply System

- ◆ Cement supply system (Surge tank) connects cement silo with the cementing mixing system to ensure the continuous and stable delivery of dry cement to the ash inlet of the high-energy mixer, reduce the dust around the mixing system, and facilitate the control of cement slurry density.



Surge tank (80-ft³)

4 x Systems of Cementing Equipment

2. Water Supply System

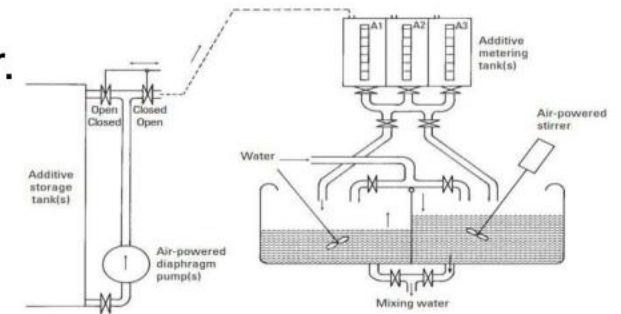
Water supply system is used to mix water and cement additives into mixing water.

◆ Type 1: LAS (Liquid-additive metering system)

LAS consists of 3 parts including storage, transportation, and metering, which can be used to continuously mixing small batches of mixing water.

Advantages:

- Requires small footprint,
- Can mix water and cement additives during the cementing process



Liquid-additive metering system
(with metering tanks)

4 x Systems of Cementing Equipments

2. Water Supply System

◆ Type 2: Mix water tank

Calculate the total volume of water and additives required for the cementing job, pour the required amount of water, and then pour each required additives into the tank and mix together.

Advantages:

- Mixing evenly.

Disadvantages:

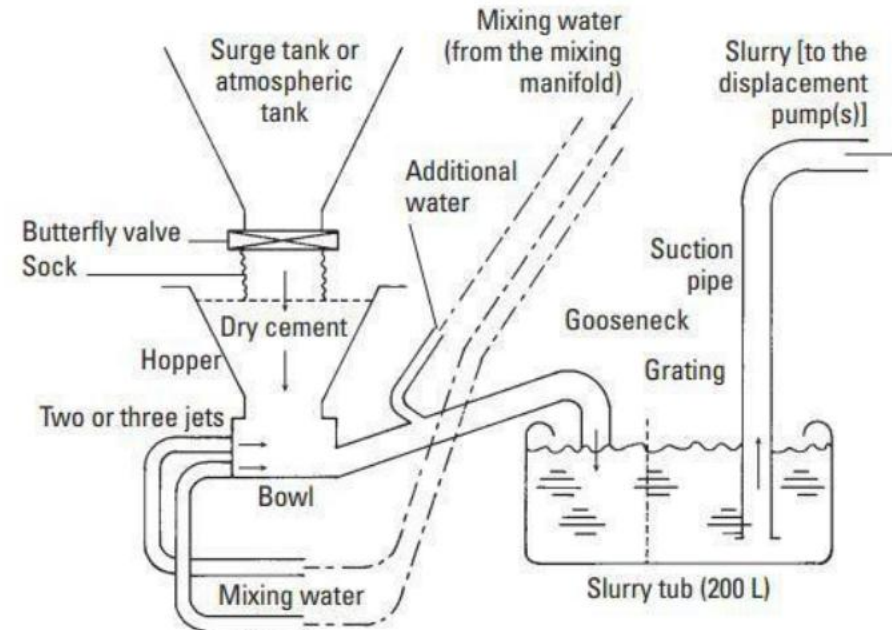
- Should mix water and additives before the cementing operation.
- Requires a large footprint and sufficient volume of mixing tanks.
- If the job is delayed or cancelled, the mixture will be wasted, and if the amount is insufficient, the normal operation will be affected.



4 x Systems of Cementing Equipments

3. Mixing System

- ◆ Mixing system is the equipments used to mix the bulk cement, mixing water into cement slurry.
- ◆ Mixing equipment is composed of high energy mixer, circulation pump, clean water pump, perfusion pump, mixing tank, stirrer, etc.



Mixing System

4 x Systems of Cementing Equipments

4. Pumping System: Pumping Cement Slurry into the Wells.

- ◆ Cementing pump is a horizontal three-cylinder single-acting plunger pump, which can be used to pump cement slurry, clean water, drilling fluid and other fluids.
- ◆ It is usually mounted on a fixed sled and powered by a diesel engine.



TPB600



TPH400

Vehicle Mounted Cementing Equipment - Onshore



Skid Cementing Equipment - Offshore



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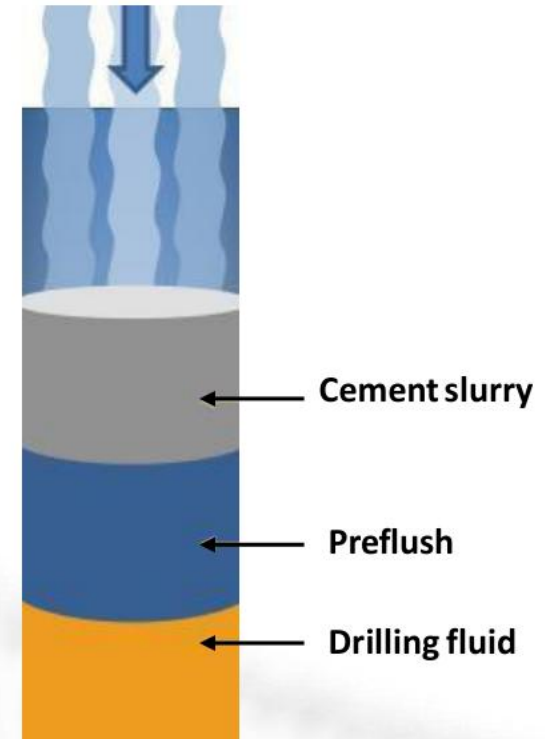
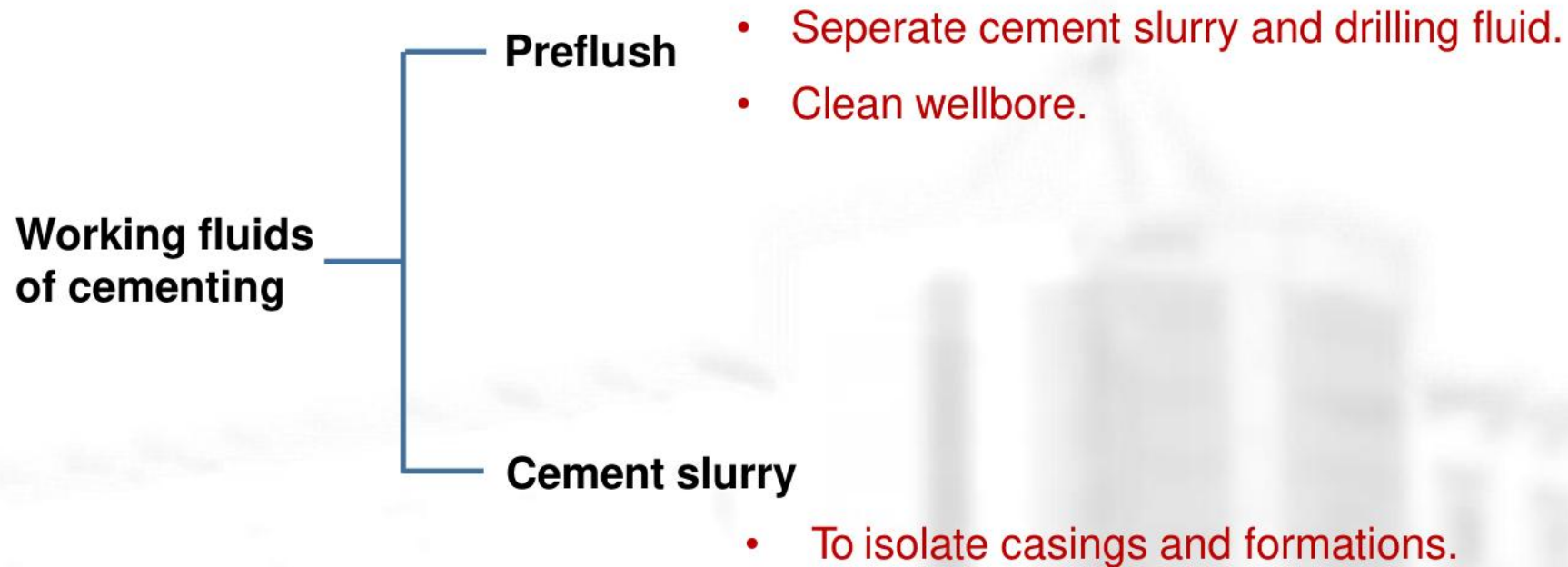
04 **SLURRY**

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06 CEMENTING CHALLENGES

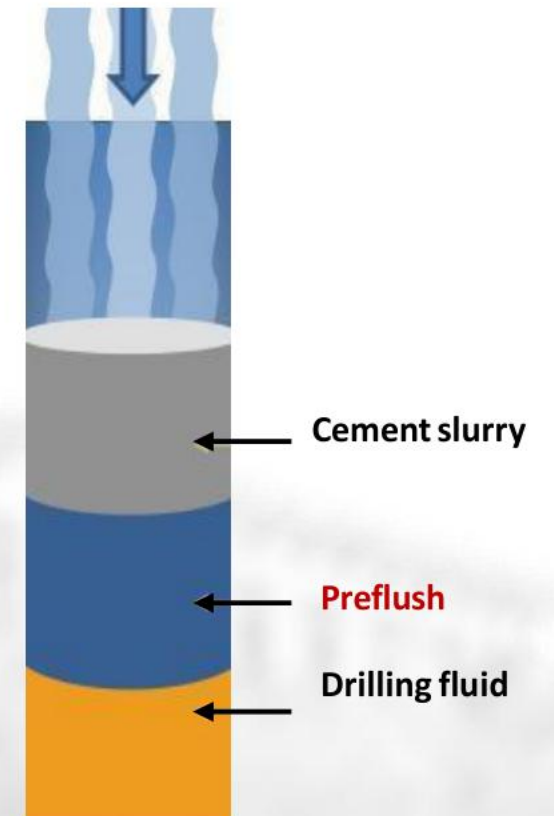


1. Working Fluids of Cementing



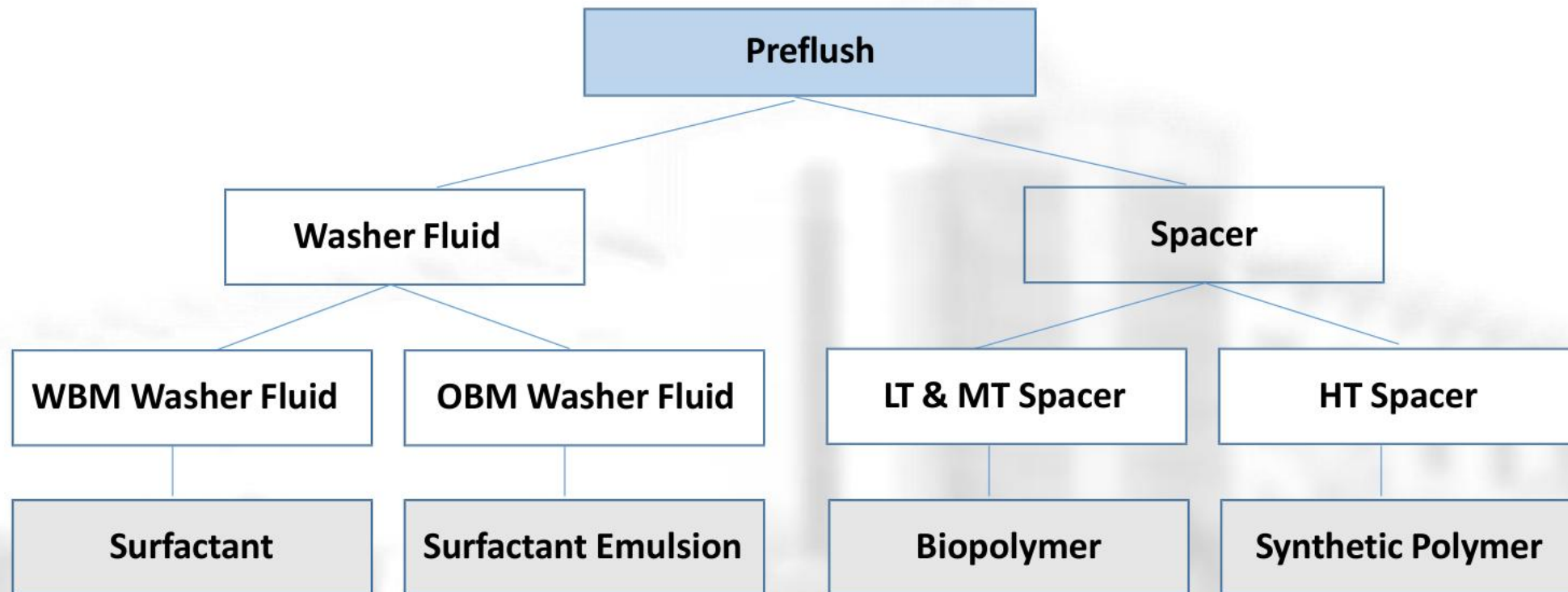
2. Preflush

- ◆ Preflush is a liquid normally pumped between drilling fluid and cement slurry.
- Effectively separate cement slurry and drilling fluid, prevent cement slurry from being contaminated by drilling fluid.
- Clean wellbore.
- Balance pore pressure.
- Replace the mud and remove the mud cake.



2. Preflush

- ◆ Flush fluid(also known as wash) can Replace the mud and remove the mud cake, increase the bonding and cementing quality.
- ◆ Spacer can effectively separate cement slurry and drilling fluid, prevent cement slurry from being contaminated by drilling fluid.



3. Cement Slurry

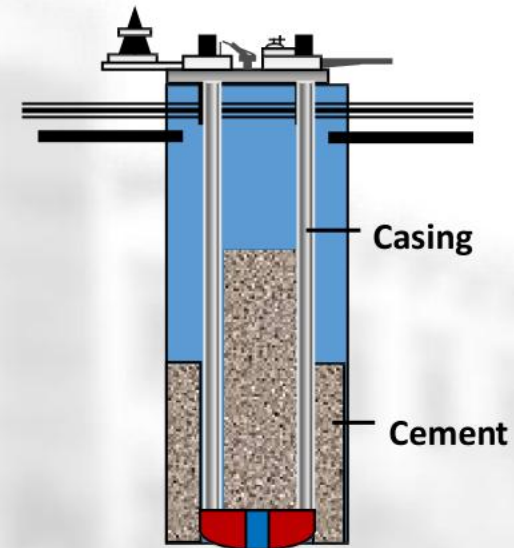
- ◆ Cement slurry is the main working fluid used in cementing operations.
- ◆ Function: Support casings, isolate casings and formations.
- ◆ Cementing is performed by pumping cement slurry from the casing into the annulus of the wellbore and casing, allowing it to return to a certain height. The cement slurry is then turned into set cement to solidify the wellbore and casing
- ◆ A cement slurry usually composed of cement, additives and water.



Cement slurry



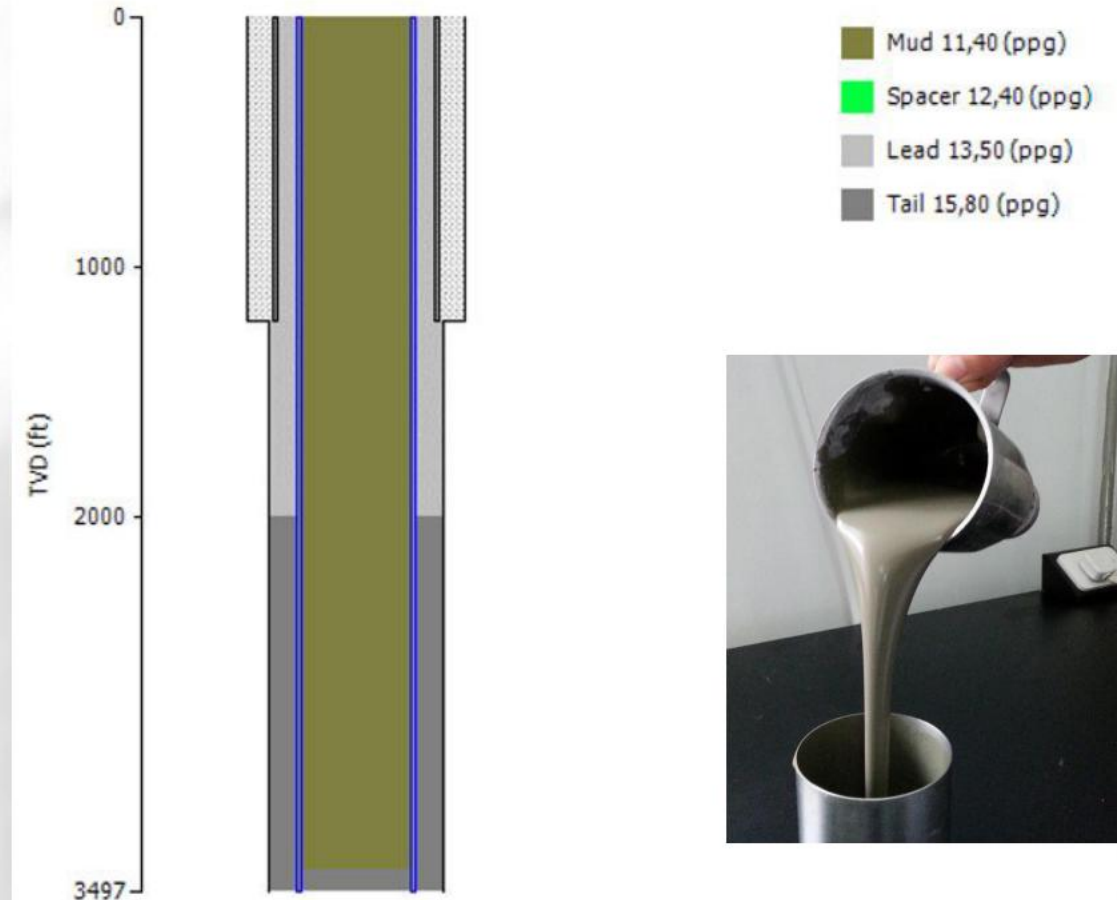
Set cement



Cement slurry in wells

3. Cement Slurry

- Lead Slurry** is a cement slurry used as the first cement to be pumped during a cementing job, typically covering the upper sections of the well. It is often lighter in density and strength than the tail slurry that follows. The primary purpose of a lead slurry is to provide zonal isolation, which means sealing off non-productive zones above the desired production zones
- Tail Slurry** is the last cement slurry pumped during primary cementing. It's typically denser than the "lead slurry" and designed to provide a strong, quick-setting seal at the bottom of the well, especially around the casing seat and targeted completion intervals. Tail slurries are crucial for isolating specific zones and ensuring well integrity



Composition of Cement Slurry -- Well Cement

- ◆ Well cement is the most important oil well bonding material. It can produce compressive strength, support casings, isolate casing and formations.
- ◆ Portland cement is the most commonly used well cement.
- ◆ Currently, there are 8 classes of API-ISO Portland cement: A, B, C, D, E, F, G, H. Class G and H cements are the mostly widely used in cementing operations.

Composition of Portland cement

Composition	Chemicals	Percentage
C_3S	$Ca_3(SiO_4)O$	40%-80%
C_2S	Ca_2SiO_4	2%-30%
C_4AF	$Ca_2Al_2O_5 \dots Ca_2Fe_2O_5$	4%-15%
C_3A	$Ca_3Al_2O_6$	3%-15%

Composition of Cement Slurry -- Additives

15 categories
+
61 chemicals



1. Fluid loss control additives
2. Retarders
3. Dispersants
4. Accelerators
5. Anti-gas migration additives
6. Light weight materials
7. Weighting materials
8. Extenders
9. Expansion additives
10. Defoamers
11. Suspension additives
12. Lost circulation materials
13. Washers
14. Spacers
15. Anti-retrogression materials

OPT cementing chemicals:

- ◆ 18+ years' development
- ◆ 15 categories + 61 chemicals
- ◆ Have been used in 5 countries and regions around the world.



■ Properties of Cement Slurry

Techincal Requirements

The relationship of cement slurry and cementing operations:

Itmes	Technical Requirements of Cement Slurry	Technical Requirements of Cementing Engineering
1	Density	Formation pressure < ECD of annulus < Fracture pressure
2	API Fluid loss	Avoid bridge plugging and formation damage
3	Rheology	Improve the displacement efficiency
4	Free water	Avoid annulus bypass
5	Thickening time	For the operation safety
6	Compressive strength	Support casing and isolation formations
7	Static Gel Strength	determine the material's ability to develop structural integrity when static, ensuring wellbore stability and preventing fluid or gas migration during operations.
8	Cement Hydration	monitor and analyze the hydration process of cement under simulated downhole conditions, including high pressure and temperature

■ Composition of Cement Slurry -- Additives

1. Improving Flow and Workability

•Dispersants:

These reduce the slurry viscosity, making it easier to pump and ensuring it flows effectively through the wellbore.

2. Controlling Fluid Loss

•Fluid Loss Control Agents:

These minimize the amount of liquid lost from the cement slurry into the formation, preventing it from becoming too thin and weak.

3. Managing Setting Time

•Accelerators:

These speed up the setting time of the cement, allowing for faster strength development, especially in cool or shallow wells.

•Retarders:

These delay the setting time, providing more time for the cement to be placed and adjusted, particularly in hot or deep wells.



■ Composition of Cement Slurry -- Additives

4. Enhancing Bonding and Strength:

•Expanding

These additives help the cement expand as it sets, filling any voids and creating a stronger bond with the casing and formation.

•Strengthening Agents:

Some additives are designed to improve the compressive strength of the cement after it has hardened.

5. Other Important Functions:

•Gas Control Agents:

These help prevent gas from migrating through the cement, ensuring a stable seal.

•Weighting Agents:

These increase the density of the slurry, which can be helpful in high-pressure or high-temperature wells.



■ Composition of Cement Slurry -- Additives

5. Other Important Functions:

•Gas Migration Control Agents:

These prevent gas from migrating through the cement, ensuring a stable seal.

•Spacers and Flushing Agents:

These are used to separate drilling fluids from the cement slurry, ensuring proper displacement and preventing contamination.



■ Composition of Cement Slurry -- Additives

Types of Cement Additives:

- **Dispersants:** Reduce slurry viscosity, improving flow.
- **Retarders:** Delay setting time, allowing more placement time.
- **Accelerators:** Speed up setting time, increasing early strength.
- **Weighting agents:** Increase slurry density.
- **Fluid loss control agents:** Reduce filtration.
- **Gas control agents:** Prevent gas migration.
- **Expanding agent:** Help cement expand as it sets.
- **Extenders:** Increase the volume of the cement slurry.
- **Other specialty additives:** For specific well conditions or applications.



Properties of Cement Slurry

1. Density

Density: used to balance the pore pressure while not fracturing the formation

Hydrostatic pressure > Fracture pressure → Lost circulation

Hydrostatic pressure < Pore pressure → Well kick or blowout

Good density design makes the whole cementing operation process safer, without well kick or losses.



Properties of Cement Slurry

2. API Fluid Loss

Good fluid loss: helps to avoid bridge plugging and formation damage.

- ◆ In small annulus cementing, high fluid loss will result in poor fluidity of slurry and increase risk of blocking the annulus.
- ◆ High fluid loss can cause formation particles to clog and formation clay to hydrate, detrimental to the reservoir.

Job type	Casing cementing	Slurry	Filtrate (mL)
Casing cementing	Intermediate casing	Lead slurry	<300
		Tail slurry	<150
	Production casing	Lead slurry	<150
		Tail slurry	<50
Remarks: Lab test under the condition of 30 min×6.9 MPa (30 min ×1000psi).			



Fluid Loss Cell



Stirred Fluid Loss Cell

Properties of Cement Slurry

2. API Fluid Loss

Good fluid loss: helps to avoid bridge plugging and formation damage.

- ◆ In small annulus cementing, high fluid loss will result in poor fluidity of slurry and increase risk of blocking the annulus.
- ◆ High fluid loss can cause formation particles to clog and formation clay to hydrate, detrimental to the reservoir.

- For slurries that dehydrate in less than 30 minutes, API Fluid Loss value is obtained by either:

$$F_{30} = 2 \times F_t \times \sqrt{\frac{30}{t}}$$

F_{30} = Filtrate at 30 minutes

F_t = Filtrate at Time

t = Time in minutes

- If slurry does not dehydrate in 30 min, then API Fluid Loss is:

$$F_{30} = 2 \times F_t$$



Fluid Loss Cell



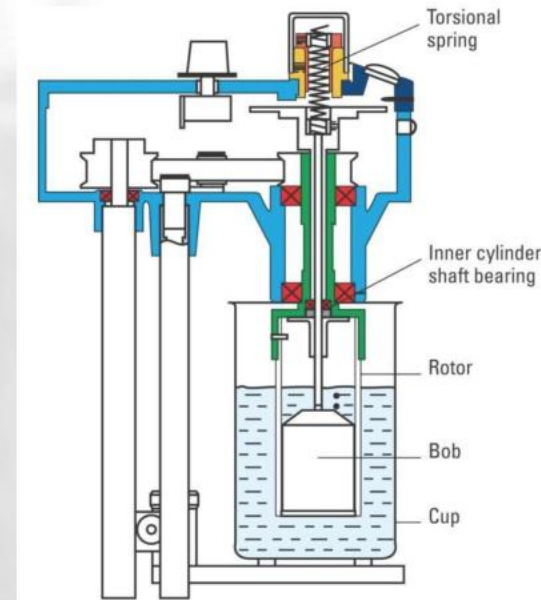
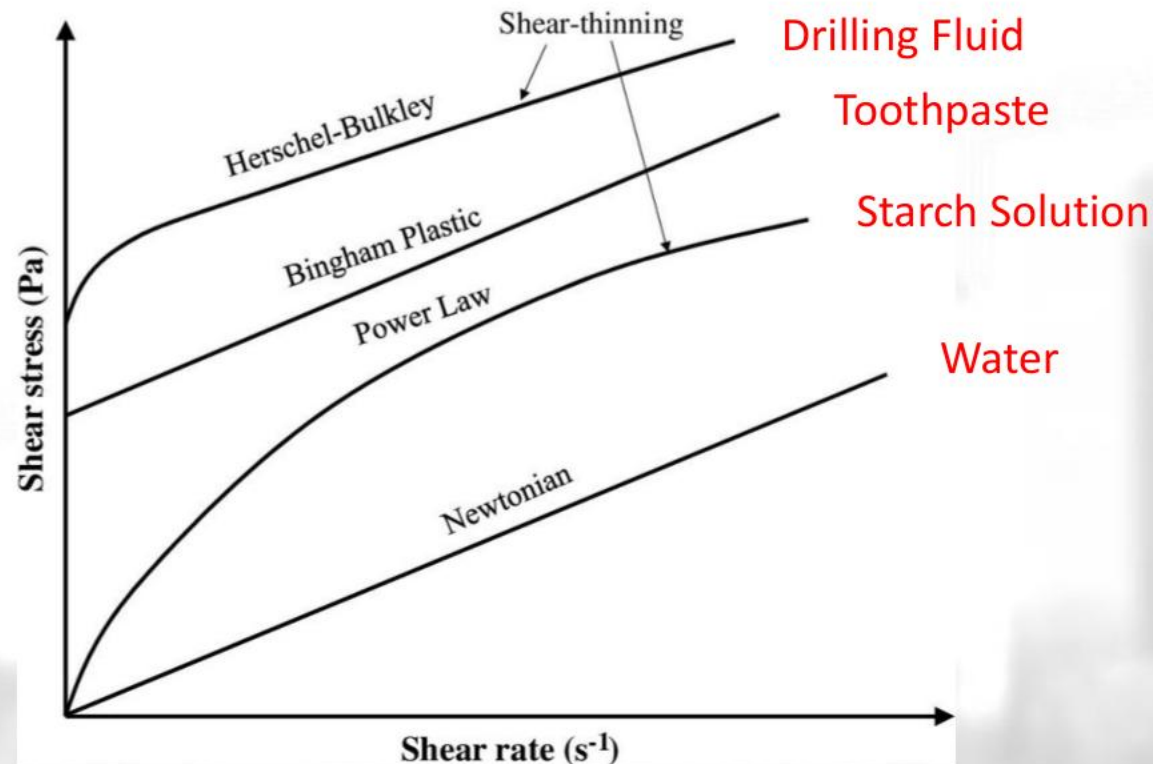
Stirred Fluid Loss Cell

Performance of Cement Slurry-

3. Rheology

Good rheology: helps to increase the displacement efficiency.

Displacement efficiency can be improved if the rheology of the cement slurry is good enough to achieve turbulent flow or plug flow.



Typical Rotational Viscometer

Performance of Cement Slurry-

3. Rheology

Good rheology: helps to increase the displacement efficiency.

Displacement efficiency can be improved if the rheology of the cement slurry is good enough to achieve turbulent flow or plug flow.

Calculated by correlation of shear stress against shear rate

Tested with Rotational Viscometer (CHAN/FANN 35)

Rotational speed (Shear rate)

300 rpm

200 rpm

100 rpm

60 rpm

30 rpm

6 rpm

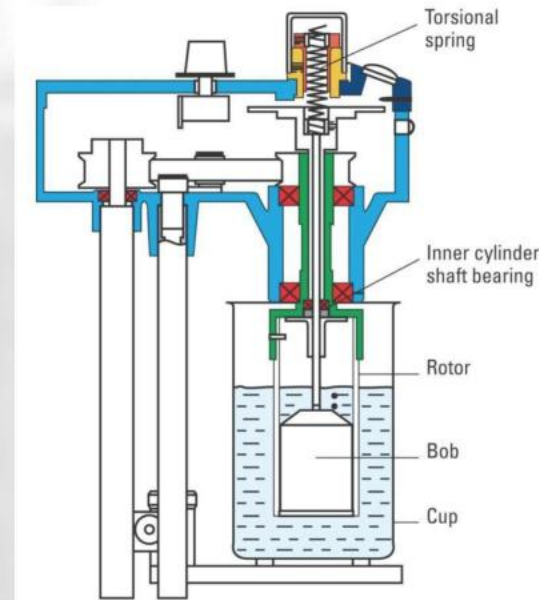
3 rpm

Approximation:

$$P_v = (\vartheta_{300} - \vartheta_{100}) \times 1.5$$

$$T_y = \vartheta_{300} - P_v$$

600 rpm not used to calculate P_v and T_y (solid and fluid will be separated)



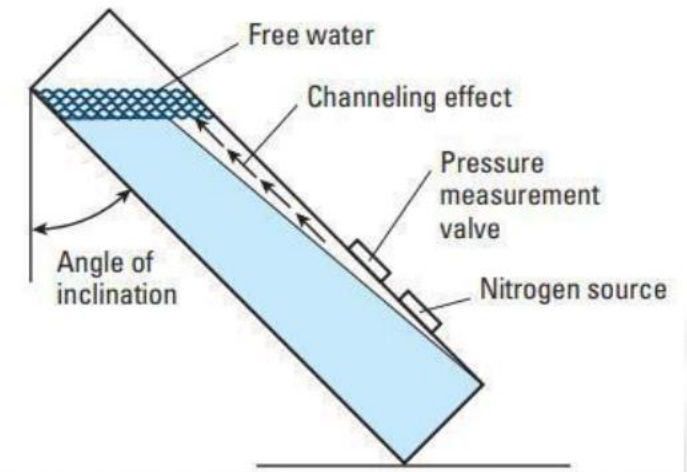
Typical Rotational Viscometer

Properties of Cement Slurry

4. Free Water

Free Water: Zero free water and good stability of cement slurry, helps to avoid cement slurry channeling and annulus bypass.

- ◆ In highly deviated Wells, excess free water can create a water strip on the upper side of the casing cement ring, connecting formations with different pressures.
- ◆ If the differential density is higher, permeability of the upper cement column can be higher.



Schematic diagram of a model showing fully developed water channeling (from Webster and Eikerts, 1979).

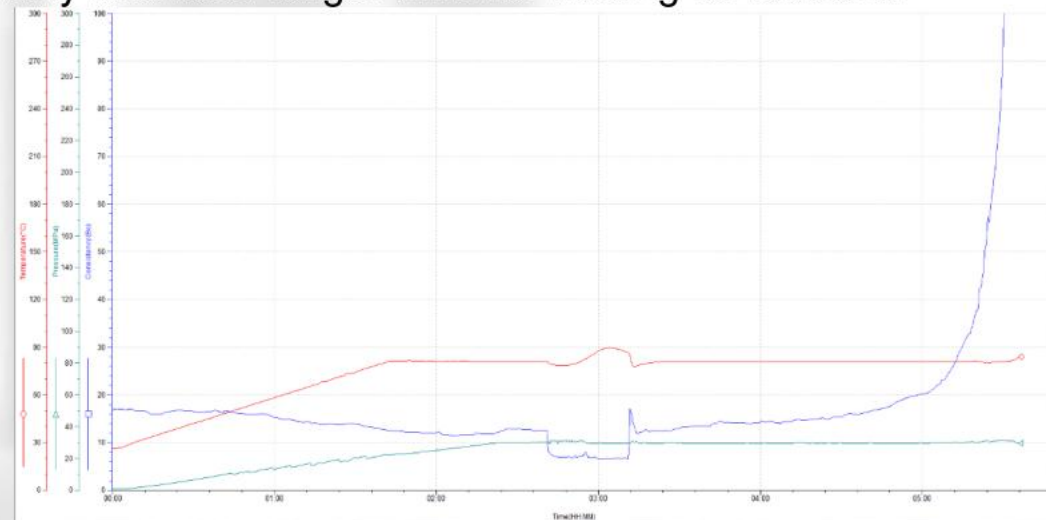
Properties of Cement Slurry

5. Thickening Time

- ◆ Thickening-time test provide an indication of the length of time that a cement slurry will remain pumpable in a well.
- ◆ The laboratory test conditions should represent the **time**, **temperature**, and **pressure** to which a cement slurry will be exposed during pumping operations.
- ◆ **Good control of thickening time**: ensure the operation safety and avoid gas migration.
 - **Short thickening time**: Unable to displace the cement slurry to the designed annulus position, leading to cement inside casing.
 - **Long thickening time**: lead to poor cement quality and also longer time for waiting on cement.



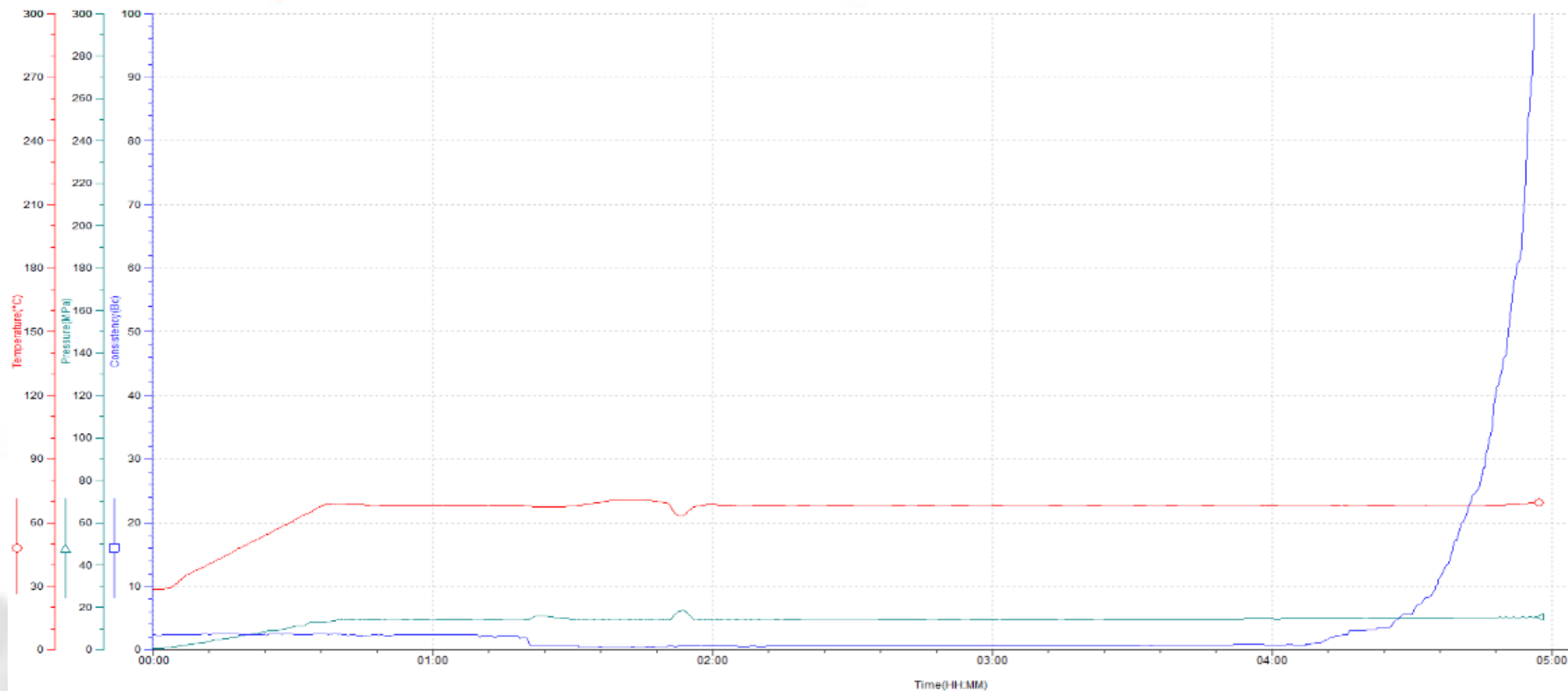
HPHT Consistometer



Thickening time curve (BHCT: 106°C)

Properties of Cement Slurry

5. Thickening Time



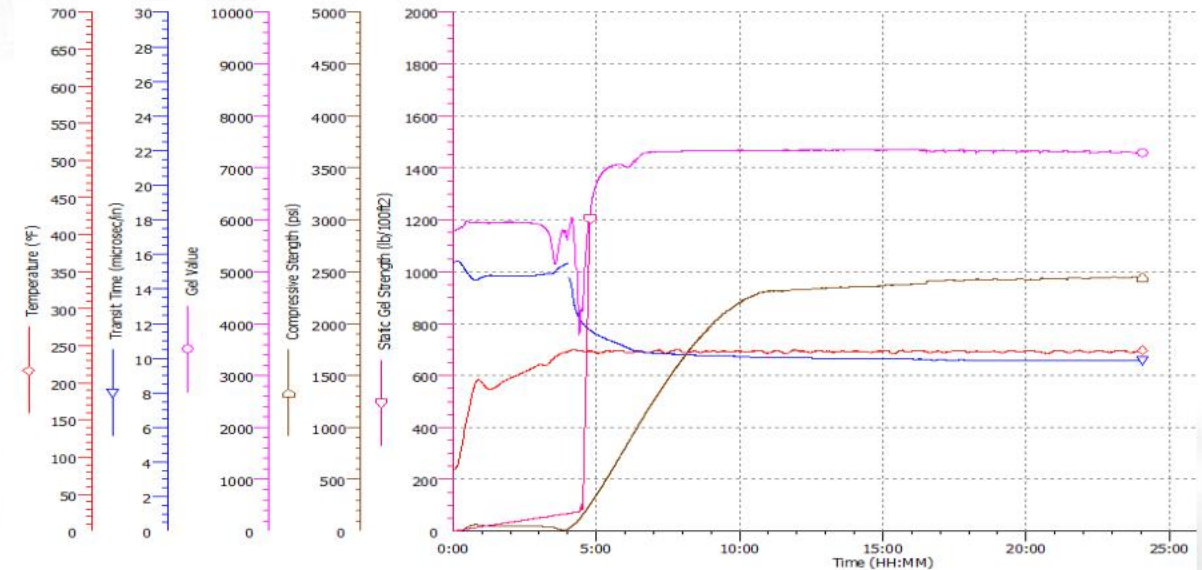
Properties of Cement Slurry

6. Compressive Strength

High compressive strength of set cement: Allow annulus cement to withstand formation pressure, support casing and isolate the formation.



Static Gel Strength Analyzer (SGSA)



Compressive strength Chart

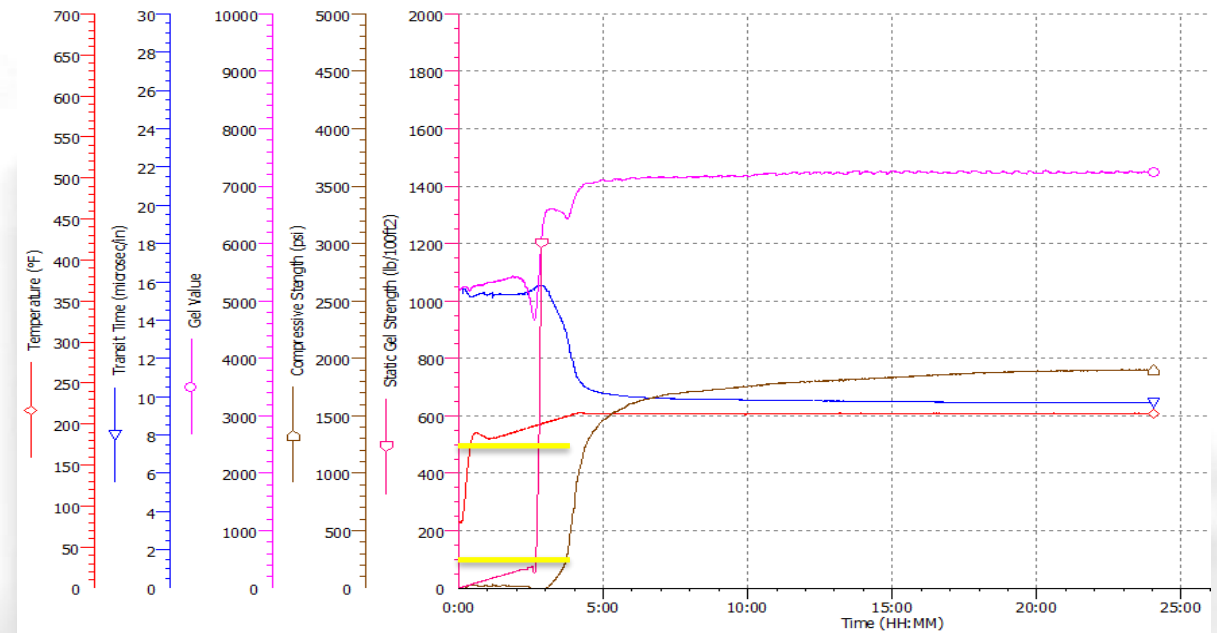
Properties of Cement Slurry

7. Static Gel Strength Analyzer

A device used to measure the development of static gel strength in cement slurries. This measurement is important in oil and gas well cementing because it helps predict and minimize gas migration, a problem where gas and fluids from the surrounding formation can invade the cement and interfere with well performance. **SGSA no. should less that 45 min (100 to 500 lb/ 100ft2).**



Static Gel Strength Analyzer (SGSA)



SGSA Chart

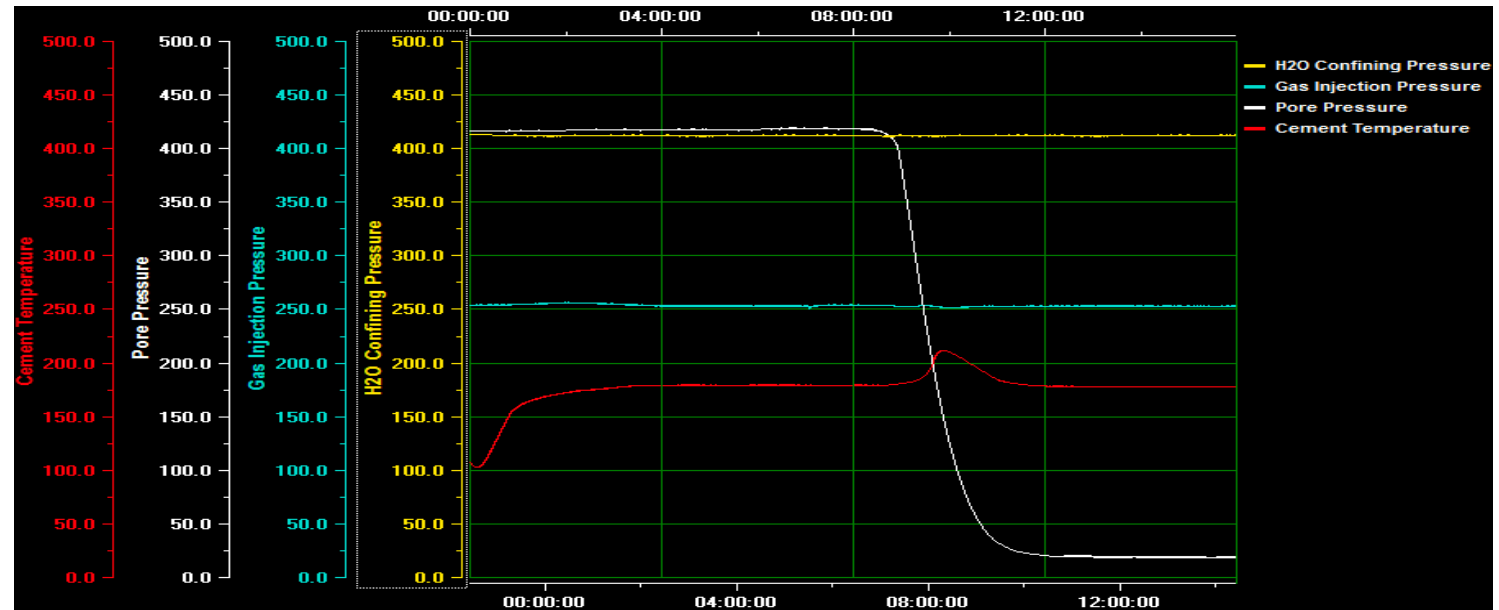
Properties of Cement Slurry

8. Cement Hydration Analyzer

Regarding **gas migration (gas penetration)**, the primary function of a **Cement Hydration Analyzer (CHA)** is to evaluate **how quickly cement develops the properties needed to resist gas invasion during the setting process.**



Cement Hydration Analyzer



CHA Chart

Summary of test procedures

Test Category	Equipment	Procedure Reference
Sampling	Diverted Flow sampler	API RP 10B-2 – Section 4
Slurry Preparation	Two speed mixer	API RP 10B-2 – Section 5
Thickening Time	Pressurized consistometer	API RP 10B-2 – Section 9
Fluid Loss	HPHP fluid loss cell	API RP 10B-2 – Section 10
Compressive Strength	Water bath, curing chamber Hydraulic press	API RP 10B-2 – Section 7
Free Fluid	250 ml graduated cylinder	API RP 10B-2 – Section 15
Permeability	Water permeameter	API RP 10B-2 – Section 11
Rheology	Rotational viscometer	API RP 10B-2 – Section 12
Static gel strength	SGSA or Vane rheometer	ISO 10426-6
Expansion & shrinkage	Annular sleeve	API RP 10B-5
Spacer/Wash/Cement Compatibility	Rotational viscometer	API RP 10B-2 – Section 16
	Reverse Emulsion Tester	CEM- Section 2B5
	Coupons, grid	
Sedimentation test		API RP 10B-2 – Section 15
Foam cement	Closed mixer cup	API RP10B-4
Specification for Cements	Various	API Spec 10A

Summary of cementing additives

PROBLEM	SLURRY PARAMETER	ADDITIVE CATEGORY SOLUTIONS
WELL CONTROL OVER PRESSURE WEAK FORMATION	DENSITY	EXTENDERS WEIGHTING AGENTS
TEMPERATURE	THICKENING TIME	ACCELERATORS RETARDERS
PERMEABLE FORMATIONS	FLUID STABILITY FLUID LOSS CONTROL	FLAC
MUD REMOVAL FRICTION PRESSURE MIXABILITY/PUMPABILITY	RHEOLOGY	DISPERSANTS GELLING AGENTS
LOST CIRCULATION	PLUGGING BRIDGING PROPERTIES DENSITY	LCM EXTENDERS
ABNORMAL AND SPECIAL CONDITION		
HEAT RETROGRESSION	HYDRATION PRODUCT	SILICA
FOAMERS	STABILIZED FOAM CAPABILITY	FOAMING AGENTS AND STABILIZERS
FOAM	FOAMING TENDENCY	ANTI-FOAM

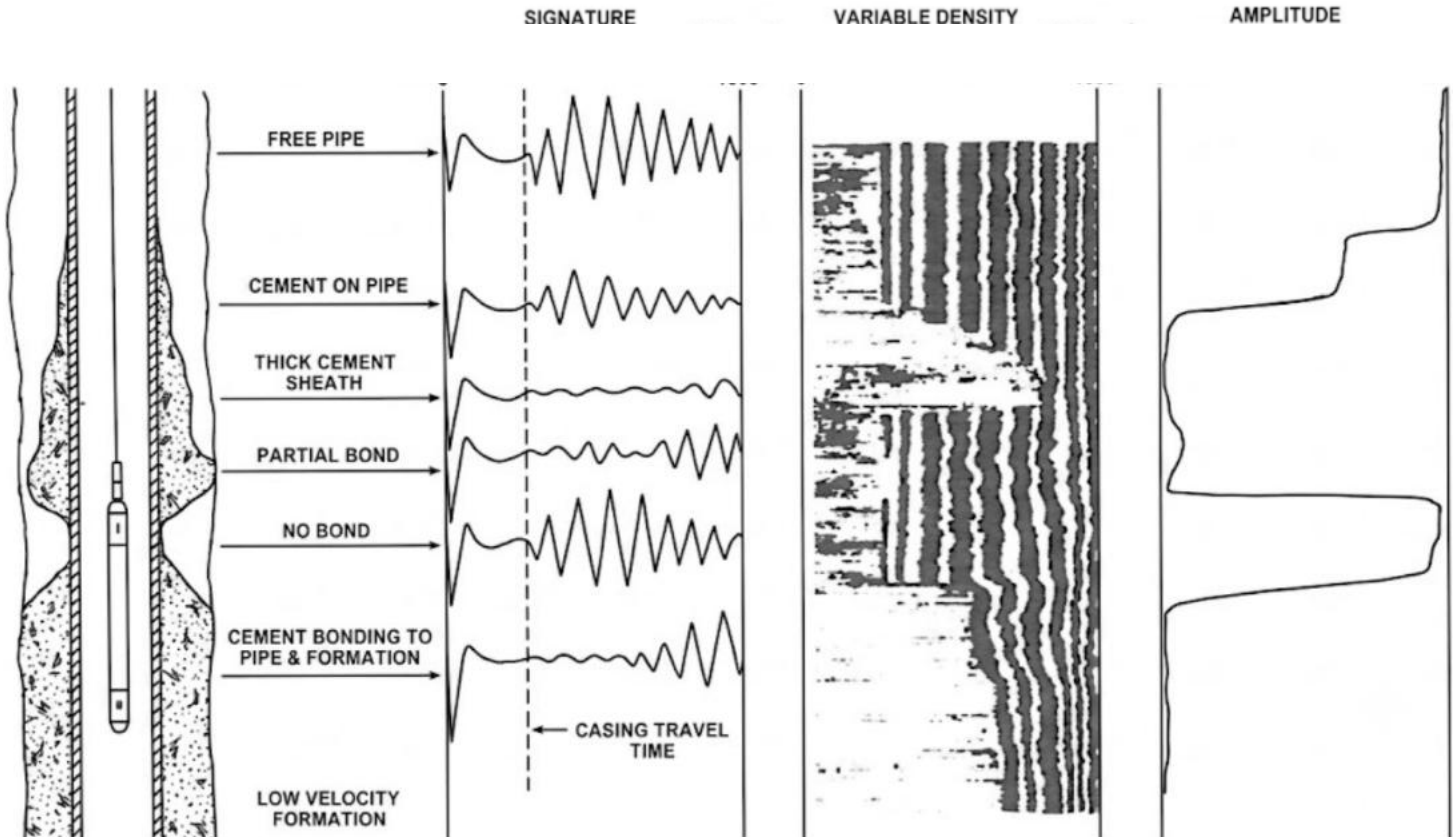
CEMENTING EVALUATION

Effects of well data inaccuracy on job success

Well Data Inaccuracy	Effect on Job Design	Effect on Job Success
Survey data	Poor casing centralization	Reduced mud removal efficiency
Caliper data	Poor casing centralization Incorrect flow technique design	Reduced mud removal efficiency
Well temperatures, formation pressures	Unacceptable slurry and set-cement properties	Excessive WOC Premature set Annular gas migration Well security problems
Mud properties	Incorrect flow technique design	Reduced mud removal efficiency

CEMENTING EVALUATION

A Cement Bond Log (CBL) determines cement integrity behind well casing by transmitting sound waves and measuring the returning signal's amplitude (loudness) and the VDL (Variable Density Log) display. Good cement absorbs sound energy; bad cement or free pipe allows the sound to bounce loudly



CEMENTING EVALUATION

1. Cement Bond Log (CBL)

Principle

Uses a sonic transmitter and receiver.
Measures the amplitude of acoustic waves traveling through the casing.

Interpretation

High amplitude

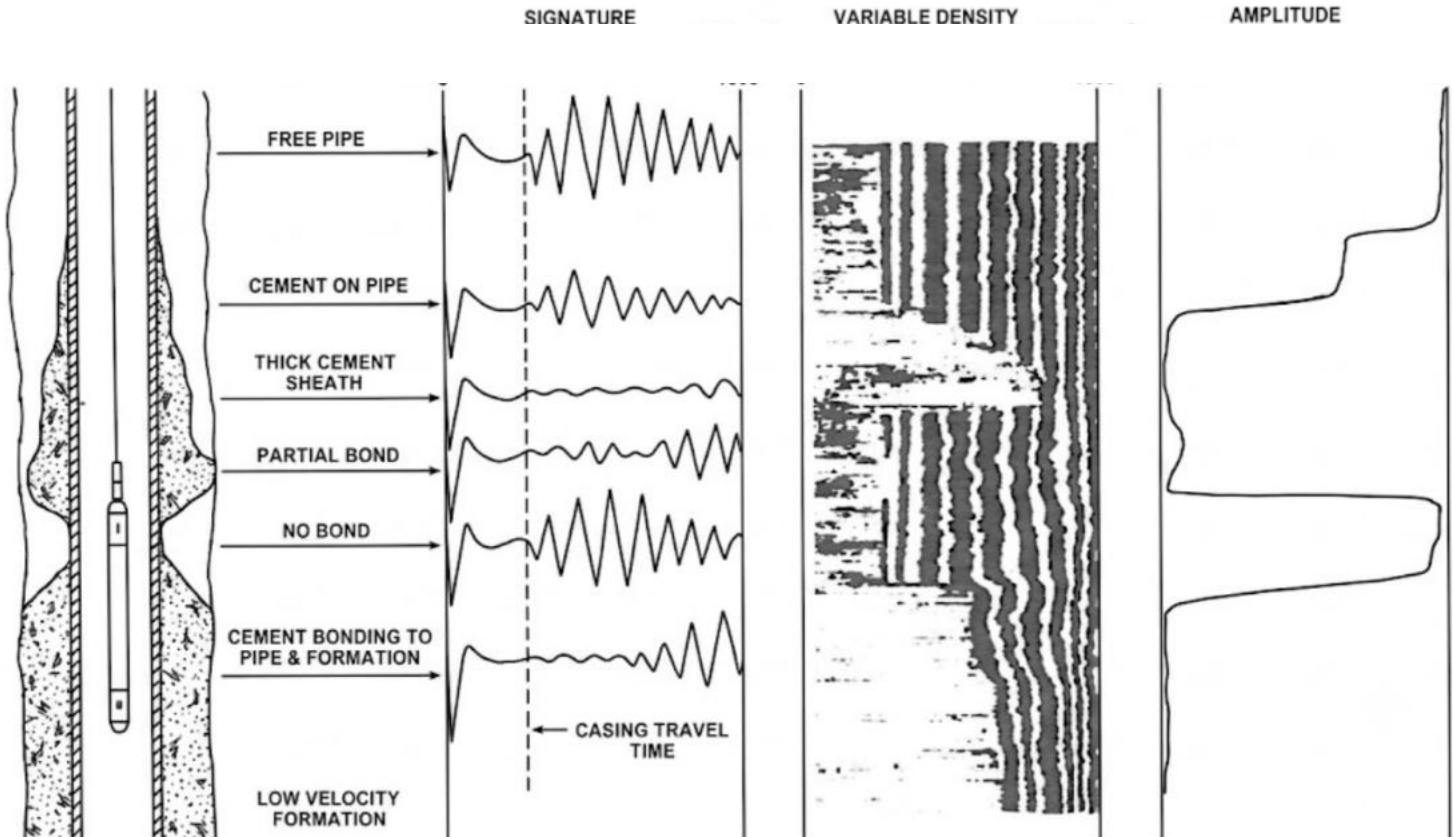
- Poor cement bond
- Free pipe
- Possible channeling

Low amplitude

- Good cement bond
- Effective zonal isolation

Output

- Bond index (0 to 100 mV)
- Amplitude curve
- Transit time



CEMENTING EVALUATION

2. Variable Density Log (VDL)

Principle

Records the complete acoustic waveform instead of only amplitude.

Typically run together with CBL.

Purpose

Detect channeling

Evaluate cement continuity

Identify free pipe

Confirm cement quality

Interpretation

Good bond:

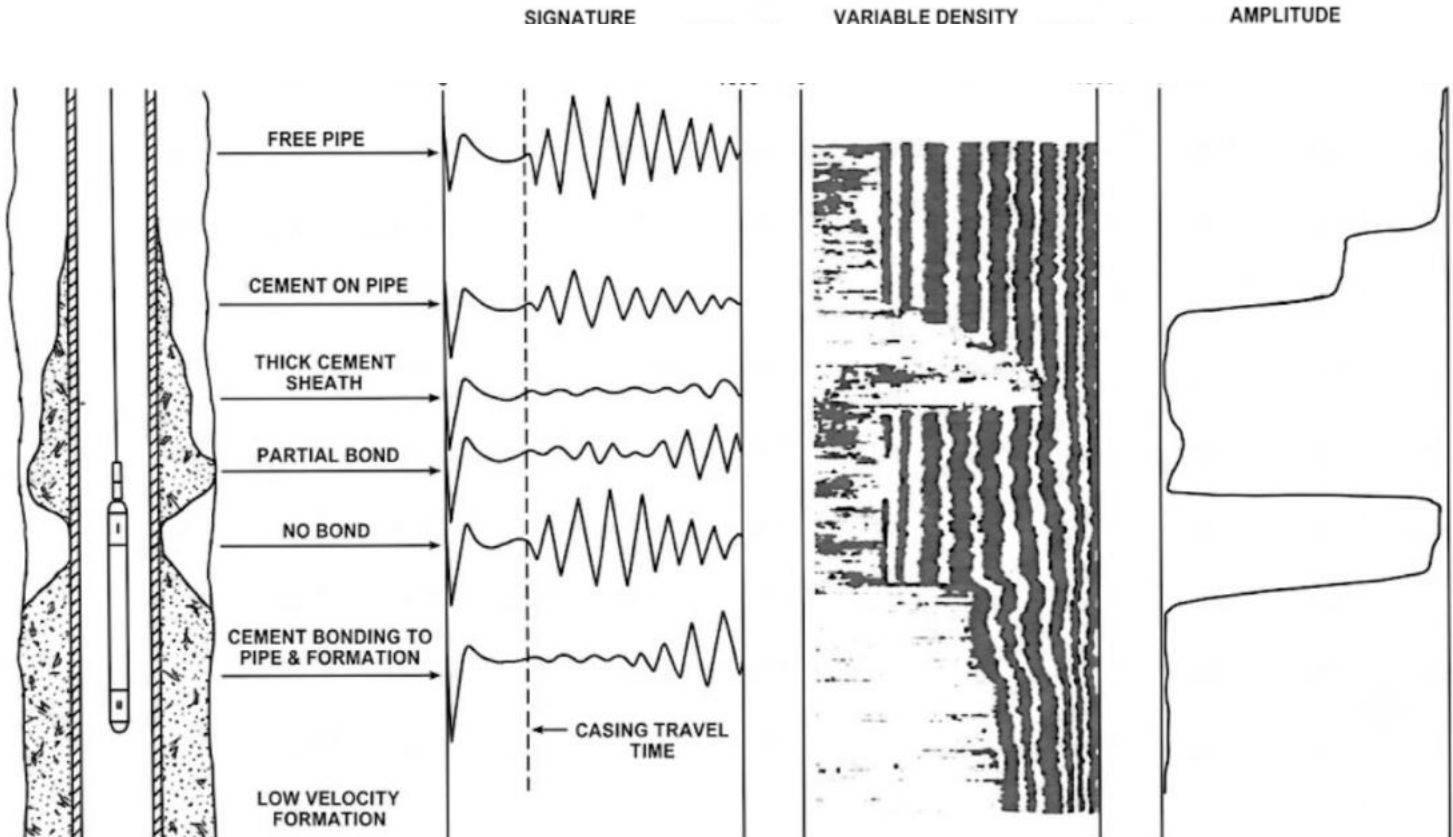
Formation arrivals are visible.

Pipe arrivals are weak.

Poor bond:

Strong pipe arrivals.

Weak or absent formation arrivals.



CEMENTING EVALUATION

3. Ultrasonic Imaging Tool (USIT)

Principle

Rotating ultrasonic transducer scans the inside of the casing.

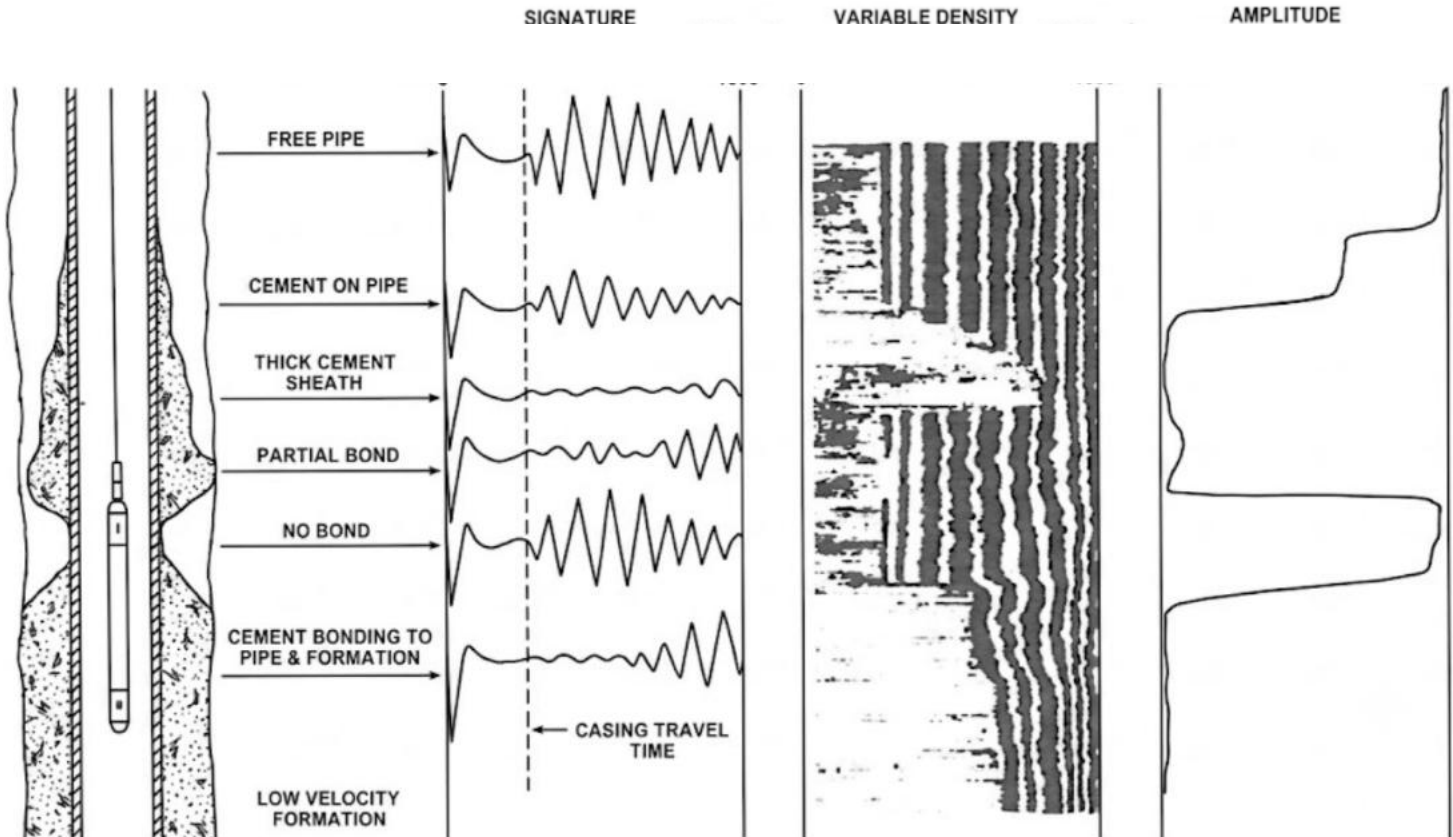
Produces a **360° image** of the cement sheath.

Measures

- Cement thickness
- Cement impedance
- Casing thickness
- Internal casing corrosion
- Micro annulus
- Gas pockets

Advantages

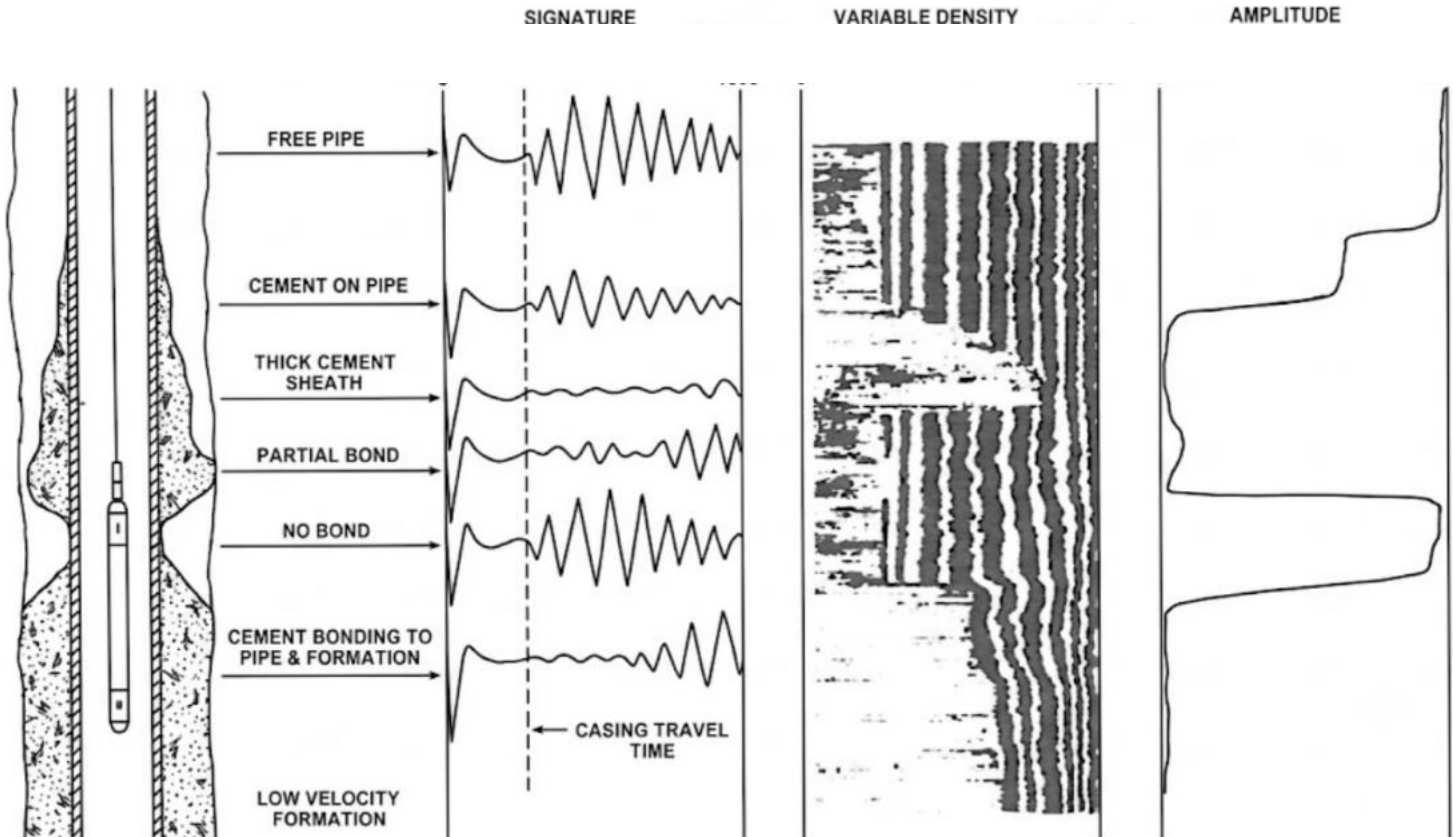
- High resolution
- Detects localized defects
- Less affected by lightweight cement than CBL



CEMENTING EVALUATION

3. Pressure Test

A **pressure test** is performed after cementing to verify that the cement and casing can safely contain the designed pressure without leaks. It is one of the most important well integrity verification steps before drilling ahead, perforating, or putting the well into production.



■ OPT Cementing Technologies

Different slurry systems are developed for different well conditions:

No.	Technology	Key Technology	Well Character	O&G	Geothermal
1	PETCEM	Unconventional Density Cement System	Lost Circulation, HPHT well		
2	EZCem	Mid to High temp cementing Technology	Robust mixing, HPHT well		
3	Flow-Off	Multifunctional Fluid migration control technology	Gas tight well		
4	FCS	Flexible cement system design for thermal and geo-stressed wells	HPHT well		
5	TOC	Thixotropic Cement Technology	Lost Circulation well		
6	OPTiWash	Special chemical wash and Spacer for OBM Displacement	OBM Drilling well		
7	LIFT	Long Interval and Fast-hydration Technology	Long interval and deep well		
8	ELVis	Effective Laminar-flow Viscous Spacer	WBM Drilling well		
9	A-CR	Anti CO ₂ & H ₂ S cement agent	Sour well well		
10	ProNET	Easily disperse LCM Fiber	Lost Circulation well		
11	SmaGEL	Smart LCM cement gel system	Lost Circulation well		

01 CONSEQUENCE A POOR CEMENTING

02 WHAT'S CEMENTING

03 CEMENTING EQUIPMENT

04 SLURRY

05 CASE STUDY

06 CEMENTING CHALLENGES



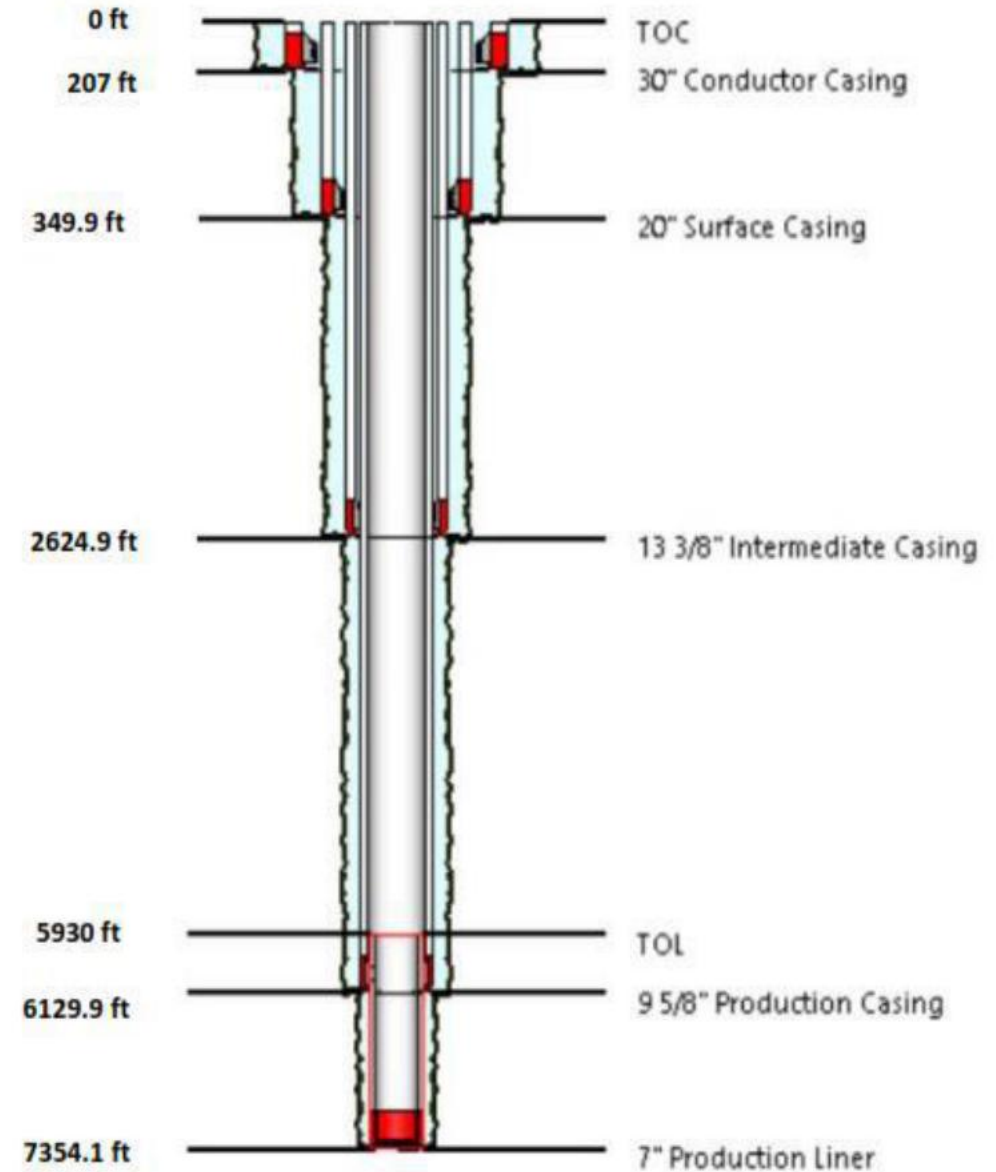
Well Information

Well Name: Manohara-01

Operation: 9-5/8" Casing Cementing

Country: Indonesia

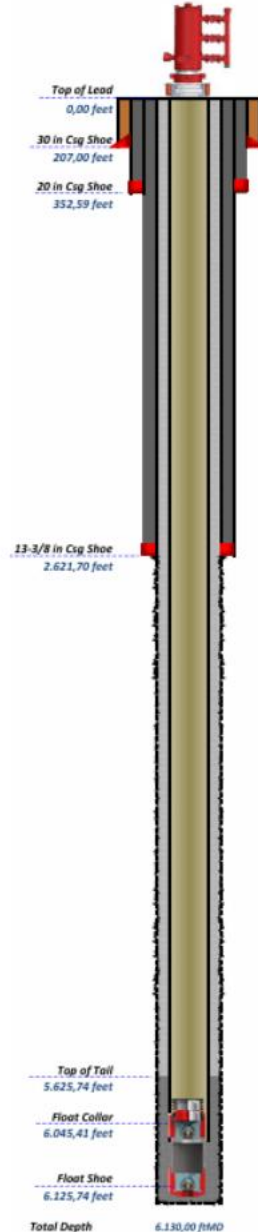
Client: KKKS



Calculation

Main Information

- ◆ Composition of slurry
- ◆ Slurry properties:
 - Density
 - Mud:10.2 ppg
 - Spacer:11.2 ppg
 - Lead slurry:12.5 ppg
 - Tail slurry: 15.8 ppg
 - Yield
 - Mixing water
 - Thickening time
 - Fluid loss
 - Free water
 - Compressive strength
- ◆ Excess of cement slurry
- ◆ Total slurry volume
- ◆ Total Preflush volume
- ◆ Total Spacer volume
- ◆ Displacement volume
- ◆ Quantity of each chemical
- ◆ Quantity of mix water
- ◆ Quantity of cement



I. WELL DATA			
OPEN HOLE	12 1/4 IN	OH EXCESS LEAD	63,00 %
FLOAT SHOE SET	6125,74 FT	OH EXCESS TAIL	63,00 %
FLOAT COLLAR	6045,41 FT	TOP OF LEAD	0,00 FT
TOTAL DEPTH (MD)	6130,00 FT	TOP OF TAIL	5625,74 FT
PREV. CASING SIZE	13 3/8 IN	CASING WEIGHT	61,00 LB/FT
PREV. CASING ID	12,515 IN		K-55 R3
PREV. CASING SET	2621,70 FT		
CURRENT CASING SIZE	9 5/8 IN	CASING WEIGHT	43,5 LB/FT
CURRENT CASING ID	8,755 IN		L-80 R3
TUBING OD		BHST	223 degF
TUBING ID		BHCT	170 degF

II. VOLUME CALCULATION						
POCKET/RAT HOLE	0,1458 BBL/FT	X	4,26 FT	=	0,62 BBL	
SHOE TRACK	0,0745 BBL/FT	X	80,33 FT	=	5,98 BBL	
CAP CSG X OH (T)	0,0558 BBL/FT	X	500,00 FT	=	27,89 BBL	
EXCESS (T)	28,51 BBL	X	63%	=	17,96 BBL	
CAP CSG X CSG (T)		X		FT =	0,00 BBL	
TOTAL VOLUME REQUIRE (T)					52,46 BBL	
TOTAL VOLUME REQUIRE (T)					294,52 CUFT	
CAP CSG X CSG (L)	0,0622 BBL/FT	X	2.621,70 FT	=	162,96 BBL	
CAP CSG X OH (L)	0,0558 BBL/FT	X	3.004,04 FT	=	167,57 BBL	
EXCESS (L)	167,57 BBL	X	63%	=	105,57 BBL	
TOTAL VOLUME REQUIRE (L)					436,10 BBL	
TOTAL VOLUME REQUIRE (L)					2.448,52 CUFT	

CASING VOLUME :	0,0745 BBL/FT	X	6045,41 FT	=	450,15 BBL	
STICKUP VOLUME :	0,0745 BBL/FT	X	8,00 FT	=	0,60 BBL	
DISPLACEMENT VOLUME :					450,74 BBL	

■ Caculations - Excess of Cement Slurry

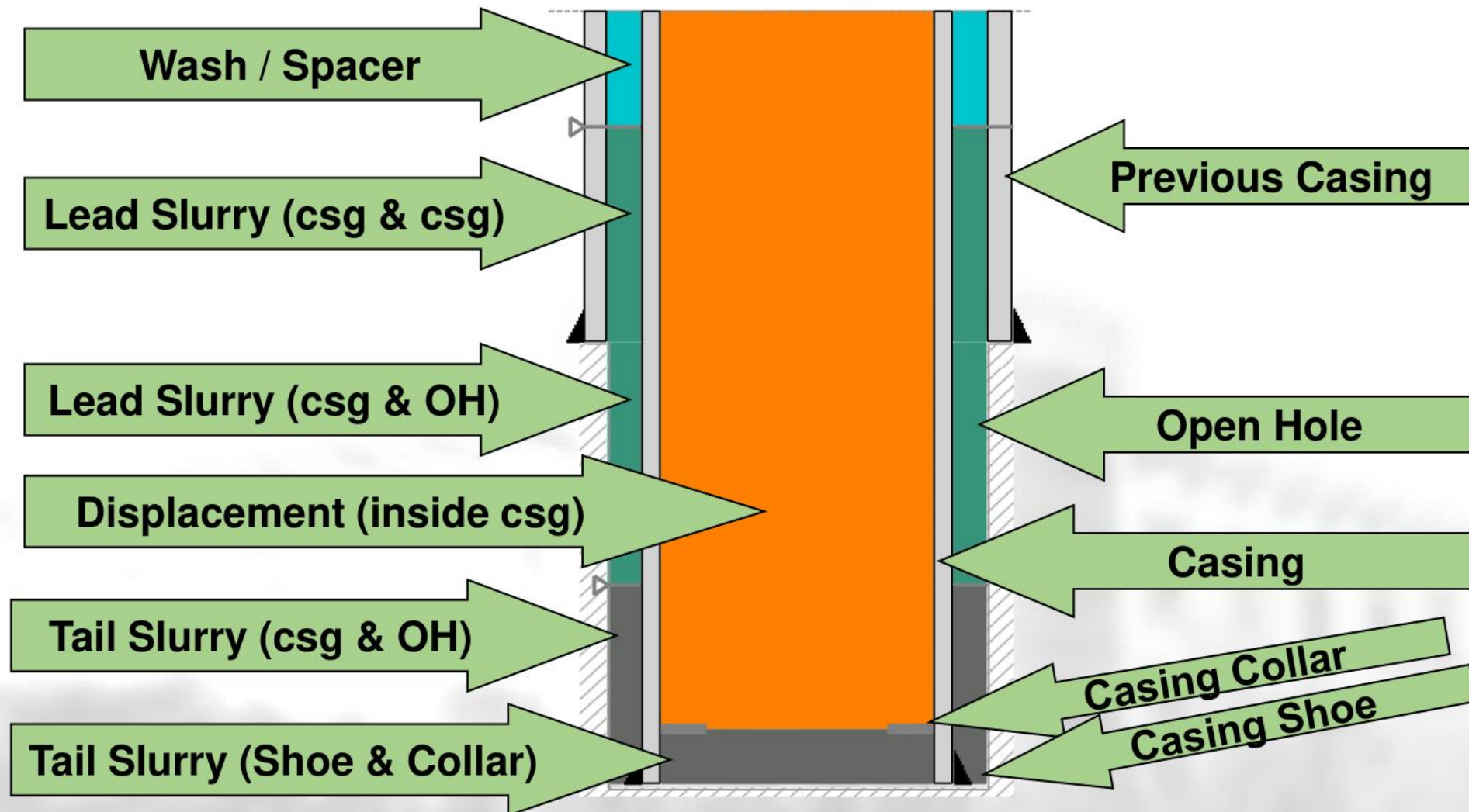
Why excess of cement slurry is needed?

- ◆ Because of the open hole irregularity or enlargement, the volume of slurry required to affect a successful cementing job is often difficult to estimate.
- ◆ In most cases, an excess allowance is made since a shortage of slurry would result in failure of the operation. The cement slurry excess is applied to the required cement volume calculated using gauge hole ID.

Rules

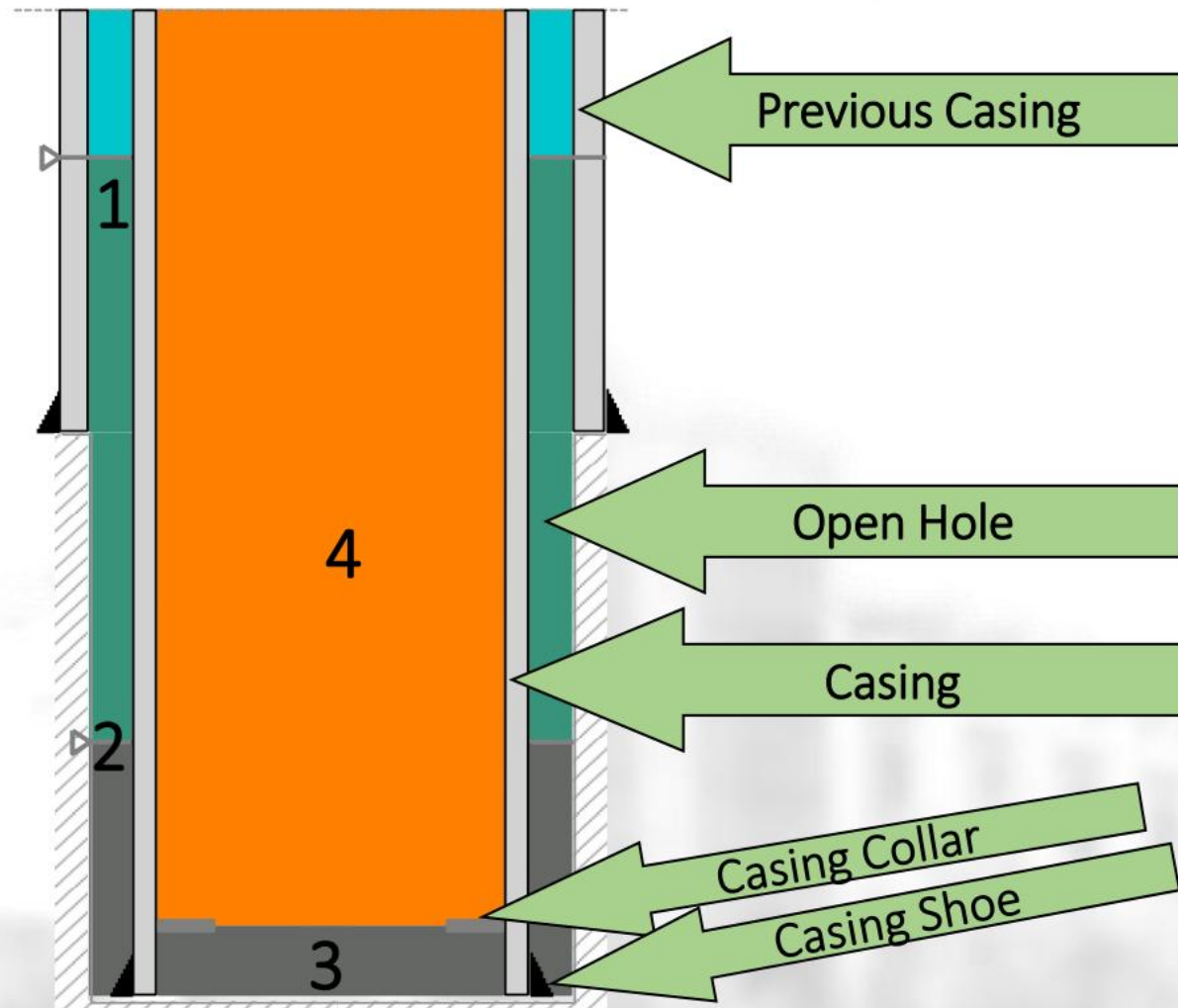
- ◆ If an open hole caliper log is available, the excess of the cement volume shall be $\geq 20\%$ of the logging annular volume.
- ◆ If the hole caliper log is not available:
 - Surface casing: $\geq 100\%$ of standard annulus volume;
 - Intermediate casing: $\geq 50\%$ of the standard annulus volume;
 - Production casing: $\geq 40\%$ of the standard annulus volume.
- ◆ For the casing overlapping sections : 0% of the standard annulus volume.

Calculations volume in single stage cementing



Calculations volume in single stage cementing

1. Betⁿ Casing & casing
2. Betⁿ Casing and open hole
3. Inside casing
(between collar
and shoe)
4. Inside casing
(Displacement volume
From surface to casing
collar).



■ Estimation cementing job time

Cementing Thickening Time Policy

The minimum Thickening Time (T.T.) of a slurry should be the greater of:

- Job pumping time X 1.5
- Job Pumping time + 2 hours.

Estimated Job Time:

The pumping time for each slurry is considered to be the time from
[when cement comes in contact with water] till [end of the job]



01 CONSEQUENCE A POOR CEMENTING

02 WHAT'S CEMENTING

03 CEMENTING EQUIPMENT

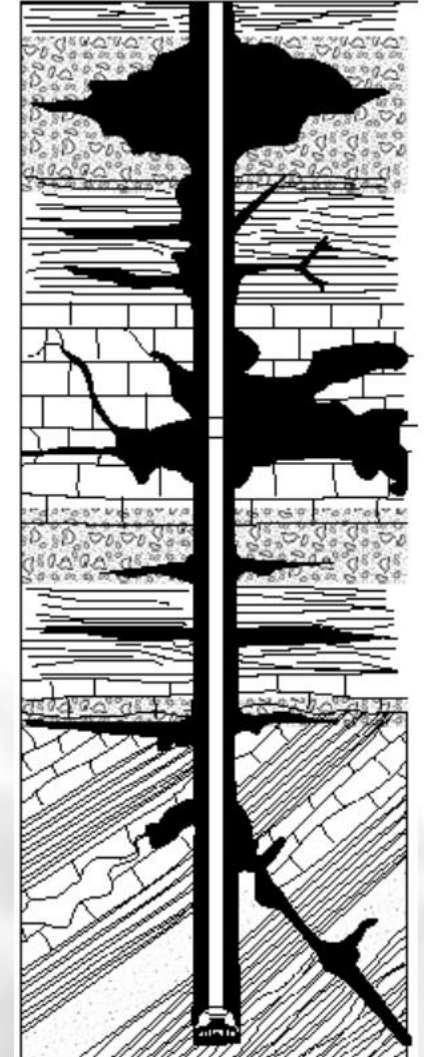
04 SLURRY

05 CASE STUDY

06 CEMENTING CHALLENGES

■ Total Circulation Losses

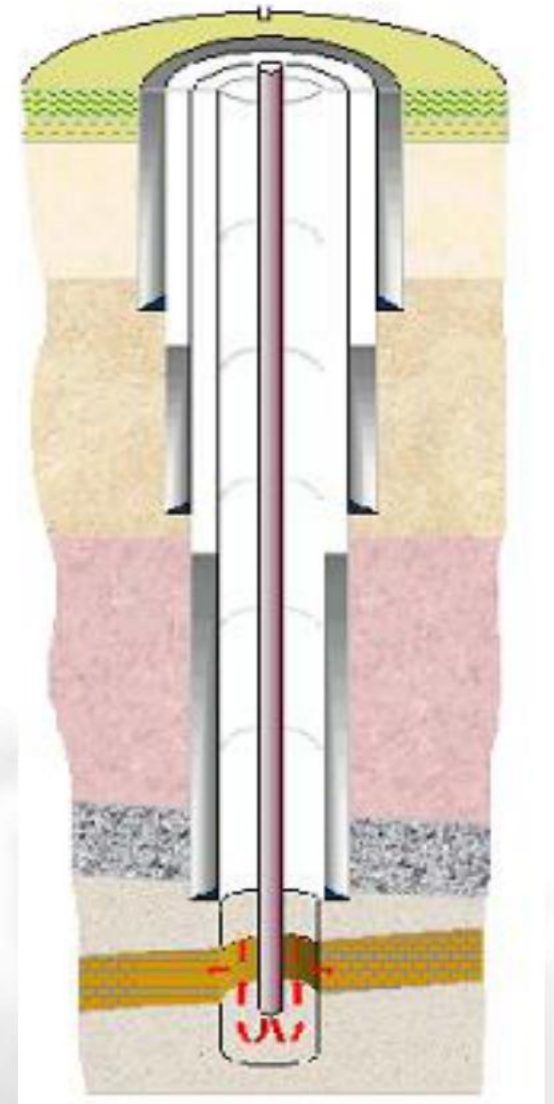
- ◆ The total or partial loss of drilling / completion fluid or cement to highly permeable zones
- ◆ Loss of a highly complex balance between pore pressure, the hydrostatic/dynamic head and the frac/parting pressure
- ◆ Uncontrolled flow of fluid in the well (e.g. mud, completion fluid or cement slurry) into a formation.
- ◆ This can occur in naturally cavernous, fissured, or coarsely permeable beds
- ◆ or can be artificially induced by hydraulically or mechanically fracturing the rock



■ Total Circulation Losses

◆ Consequences:

- **Time consuming**
 - Lost time due to slow drilling progress
 - Poor cuttings removal (stuck pipe)
 - Additional casing strings
- **Missing the target**
 - Damage to pay zone resulting in skin
 - Poor zonal isolation due to cement losses
- **Dangerous operation**
 - Kick/Blow out due to imbalance in the well
 - Loss of well



SmaGEL

- Gelling Block cements is the property of becoming fluid when in motion, such as when being pumped and of forming a gel when allowed to stand. Thus, KCM038 slurries are thin when mixed and pumped, but after motion has been stopped, they rapidly form a gel structure. This gel structure is strong enough to support the hydrostatic pressure, preventing the fallback. If sufficient force is applied to move the slurry, the gel structure is disturbed, and the slurry returns to liquid phase and pumpable state. Gel structure will support the weight of slurry column and preventing fall back.
- Gelling Block systems develop gel strength according to the additive concentration. Reduced gel-formation time and increased gel strength can be achieved by adjusting the KCM038 concentration. Laboratory tests show that KCM038 provides uniform gel strength properties when used with a widely various of cement brands.
- KCM038 slurries do not expand upon setting and remain dimensionally stable. It can be also used with any Portland cement. Another advantage of using KCM038 is that higher early and ultimate compressive strengths result.



■ SmaGEL

◆ Benefit:

- Manages circulation losses in cavernous or vuggy formations where bridging materials and LCM are ineffective.
- Minimizes the risk of premature setting of cement in case of pumping interruption by imparting mild gells up properties.
- Increases early and properly compressive strength as compared with different conventional gypsum-based thixotropic additives.
- Simplifies cement design are recommended for this system.

◆ Treatment:

- Gelling Block cements slurries are designed for use at the temperatures: 80 - 185°F (27-85°C) BHCT. Need another material if we want apply in high temperature. (Recommend to use High Temp TOC Agent).
- When the slurry density is between 1.5-1.7 SG, 0.3 - 1.5 gps is considered optimum concentration in most fluid design.
- When meet the loss circulation condition, Recommend to use density 14.5 – 15.0 ppg because if we use too high density will be impacted to viscosity and problem during mixing.
- Density more than 15.0 ppg only for stabilization plug.

GEOTHERMAL SmaGEL

Lab Test Video & Documentation

Mixing Process



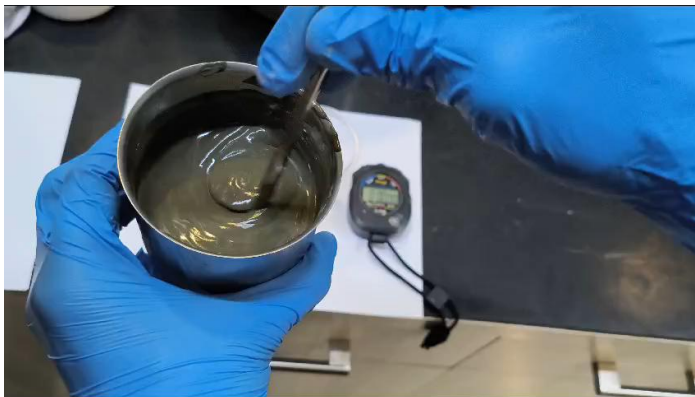
After Mixing



Measuring Rheology



After Conditioning



After 25 Min Static

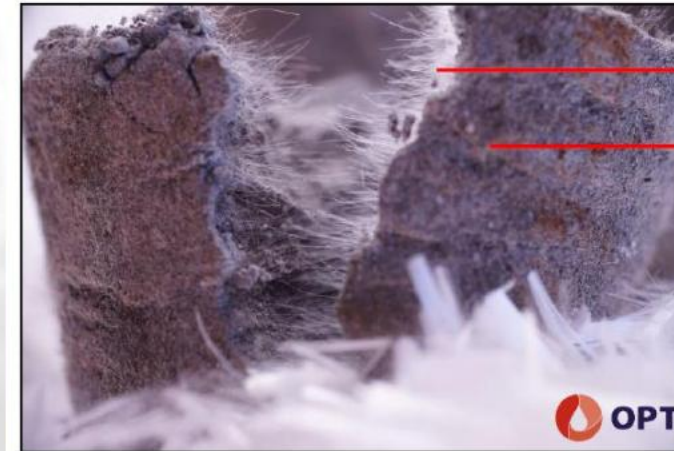


Re-Mixing After Static



ProNET

- ✓ ProNET™ technology is advanced fiber materials which can link each other and mix with cement to seal lost circulation. The cement particle can help to form bridging against fiber, the more cement solid content in slurry system, the more bridging effect will form to isolate the lost circulation zone
- ✓ Advanced fibers mixed with cement to seal lost circulation
- ✓ Not recommended to treat losses while drilling by adding to drilling fluids.
- ✓ Engineered to an optimal size
- ✓ Inert
 - No effect on Thickening Time
 - No effect on compressive strength development
 - Compatible with all cementing additives
- ✓ Extremely effective with PETCem* systems
 - Better plugging effect due to higher solid content (SVF)
 - Lower rheology

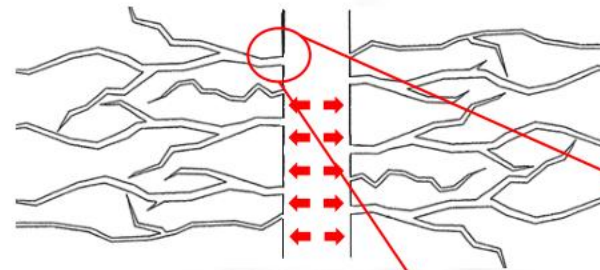


Fibrous Material/ ProNET*

Light Weight Cement/ PETCem-LD

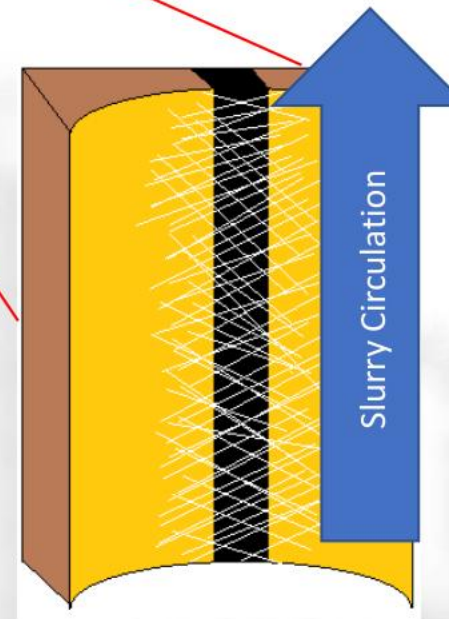
ProNET

ProNET* Mechanism Work



Lost Circulation Zone

High Solid Volume Fraction Slurry (PETCem-LD) Combined with Fibrous material (ProNET*) form the powerful nets on the fracture opening zones to plug lost circulation area and keep cement slurry circulation.



REMEMBER

- **Good Operations with Fatal Accident is Nothing**
- **Start the Job is NOT step Number 1**
- **Design – Preparation – Execution – Evaluation**

**THINK
WORK
STAY** **SAFE**

Teamwork - Innovation - Excellence



THANK YOU

"THINK SAFE. WORK SAFE. STAY SAFE."

